Theory Book

COMPUTER APPLICATIONS TECHNOLOGY

12

basic education

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GENERAL CONCEPTS

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Unit 1.1	Computers and their uses
Unit 1.2	Data, information, knowledge and wisdom
Unit 1.3	Convergence
Unit 1.4	Social implications: Environmental

By the end of this chapter, you will be able to:

- Describe the various reasons for using computers.
- Describe the role and use of data, information, knowledge and wisdom as part of information management.
- Explain convergence.
- Explain the social implications of computer technology on the environment.

INTRODUCTION

To get a better understanding of the topics that will be discussed in this chapter, we will start with a quick review of what Information and Communications Technology (also known as ICT) is and how it is related to computers.

The term ICT refers to technology that gives access to information by using **telecommunication**. The information is obtained from either a telephone or a computer network. Users can access this information by using a computer, or a computing device, such as a smartphone. ICT also includes the most common and widely used network – i.e. the internet. Thanks to advances in technology, users can access the internet via desktop computers, laptops, tablets and smartphones.

In Grades 10 and 11, you learned about different types of computers, such as the following:

- **Server:** A server is a computer that has powerful processors, large hard drives and plenty of memory power. They are used in networks where large amounts of data need to be stored so that the computers on this network can access the data. Servers also make it possible for computers on the same network to share other devices, such as printers.
- **Workstation:** A workstation is a computer intended for individual use that is faster and more capable than a personal computer. It's intended for business or professional use (rather than home or recreational use). Workstations and applications designed for them are used by small engineering companies, architects and graphic designers.
- **Personal computers (microcomputers)/Desktop:** A personal computer, or more commonly known as a PC, was commonly referred to as a "microcomputer", because, when compared to the large systems that most businesses use, it is a compact computer with a complete system. They are the smallest, least expensive and most used type of computer. They are physically smaller, have a relatively small memory, have less processing power, and permit fewer peripherals than super and mainframe computers. Desktop PCs are *not* designed to be carried around because they are made up of separate components.
- **Laptops/Notebooks:** Laptops are also known as "notebooks". They are portable PCs that combine the display, keyboard, processor, memory, hard drive and cursor positioning device (a touchpad or trackpad) all in one package. Laptops are battery-operated and as a result, are completely portable.
- **Tablets:** Tablets are smaller than normal laptops and are ultra-portable (easy to carry). They are generally cheaper than brand new laptops, and their processors and other components are less powerful than that of regular laptops.
- **Smartphones:** Handheld-sized computers that use flash memory instead of a hard drive for storage. They have virtual keyboards and use touch-screen technology. Smartphones are lightweight and have a good battery life. (The battery life of smartphones varies, depending on the make.)
- **Embedded systems:** Embedded systems, or dedicated devices, are stand-alone electronic hardware that is designed to perform dedicated computing tasks, for example automatic teller machines (ATMs), MP3 players and so on.

In general, computers have certain economic benefits – for example, they save paper, labour, communication speed and cost, and so on.

UNIT

1.1 Computers and their uses

Throughout history, there have been many inventions and discoveries that have changed the way in which we live our lives. One such invention is the computer.

Computers have enabled us to make technological and scientific advancements, such as exploring the deepest depths of the oceans and outer space. Most of all, however, they allow us to stay connected with people all over the world.

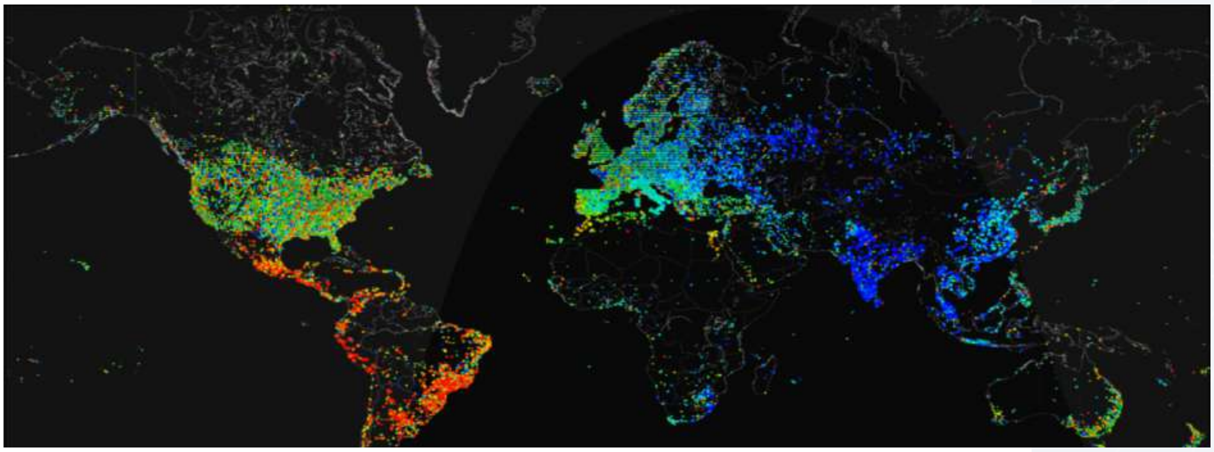

Figure 1.1: *Internet usage across the world over a 24-hour period*

In today's world, most people living in a city will interact with a computer in one way or another and on a daily basis. This can be a simple interaction, such as stopping at a traffic light, or a more direct interaction, such as using a smartphone or laptop. One thing is certain, however: Without computers, our lives would be very different!

WHAT IS A COMPUTER AND HOW DOES IT WORK?

All computers, whether they are the smartphone in your hand or large, powerful servers, operate on the same five basic principles. These are input, processing, storage, output and communication. Each component of a computer performs one of these functions, but they all work together to make the computer work.

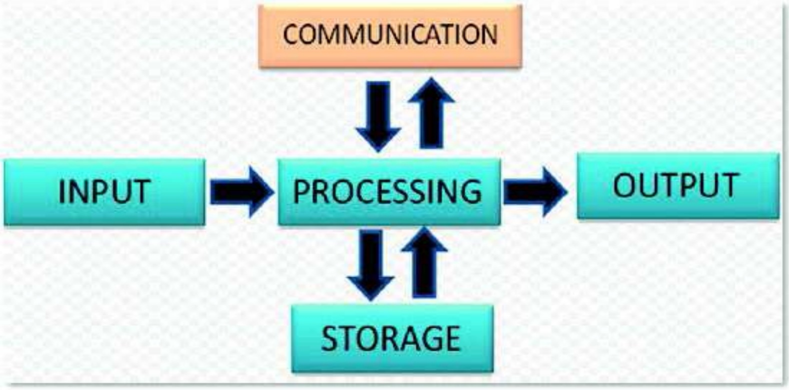

Figure 1.2: *The stages of the information processing cycle*

In this section we will look at each of these stages and how they work together. We will also look at how these processes can get one computer to communicate with users and other computers. The five main steps are input, processing, storage, output and communication.

INPUT

In the input stage, the data is entered into the computer. There are many ways to do this. In fact, there are as many ways to input data as there are input devices. You would have learned about input devices in Grade 10 but just to refresh your memory, input devices are things such as keyboards, touchscreens and microphones. The user inputs the data (for example, by typing on a keyboard or speaking into a microphone) into the computer. The device takes this data and converts it into a series of 1s and 0s (this is called binary code).

PROCESSING

The central processing unit (CPU) inside the computer then takes that binary code and does the calculations needed to get that data to display in a way that makes sense to the user. The CPU works with the computer's memory to get instructions on how to display the information from the input device and stores it as pixels in the computer's memory. This information is sent to the output device to be translated and displayed in a way that is useful. All of this takes a fraction of a second to do.

STORAGE

Storage is where the computer takes the input and stores it in its memory banks. There are many ways to store the data, but the basic process is as follows:

1. The CPU writes the data to the computer's temporary storage, or random access memory (RAM).
2. The computer then waits for the user's command to move the data from the RAM to more permanent storage. If that command is given, the computer writes the data to the disk drive.
3. Lastly, the computer saves the data in a location on the drive, either the default storage location or a location set by the user. The user can then recall this stored information at any time.

You can also store information using external storage devices (for example USB drives or external hard drives).

OUTPUT

Output is where the computer takes the pixels from the processing stage and displays them in a way that the user can see them. There are many kinds of output devices, such as printers, screens, video and audio devices.

These devices make the raw data usable and visible, allowing human users to interpret the data, turning it into information. This could be the sound waves of a song or the letters in a document.

WHY DO WE USE COMPUTERS?

Computers play a big role in our daily lives, because they can do the following:

- Help improve productivity
- Assist scientists to cure disease
- Help architects design and construct intricate new buildings
- Empower people from poor countries by opening opportunities across the world

In the following sections, we will take a brief look at some of the reasons why computers are used.

SAVING TIME

One of the major benefits of modern computers is that they can save us a lot of time and effort – from finding the quickest route to the mall using Google Maps, to sending an urgent email to a work colleague. Each activity is completed much faster and with much more ease; all thanks to the computer.

Let's take a look at some of the ways in which computer technology helps to save time:

- You can use an online shopping website so that you can do your grocery shopping from the convenience of your own home and have the groceries delivered to your home. By doing so, you save the time it would have taken to drive to the shop, do your shopping, drive back home and unpack your groceries from your car.
- You no longer need to go to your bank to do transactions. Instead, you can use your bank's online banking facilities to view your bank balance, pay your bills, or transfer money.
- Instead of standing in the queue at your favourite take-away restaurant, you can order your food from the restaurant's website.
- You can view online traffic cameras and maps with traffic information to find the quickest route to a specific location.
- You can do certain tasks, for example difficult calculations, much faster than if you had to do them manually.
- You can find information quickly by searching on the internet, or on a **database**.

Something to know

Computers can complete tasks that are impossible or incredibly time consuming for humans to do.
For example, in 1624, Henry Briggs published a book containing the logarithms for 35 000 numbers that took him years to calculate. Today, a person with a computer could do the same work in less than five minutes!

COMMUNICATION COSTS

Computers have greatly reduced the costs of communication with people across the world. Video conferencing is much cheaper that buying a plane ticket and flying to a meeting, and sending or sharing files over the internet.

EFFICIENCY

Computers made it possible to obtain, store and record data both quickly and efficiently. For example, you can research any topic on the internet in less than an hour. Many repetitive tasks performed by humans can be time consuming and there is always the risk of human error. Using computer technology can reduce the time it takes to complete the task and it can reduce or eliminate mistakes.

SAVING LABOUR

The automotive industry is an example of how computers and **automation** can help with saving labour. It would take a person roughly two months to two years to assemble a car by hand. This is significantly longer that it takes when using computers and automation, which can produce a fully assembled and painted car in about 8 hours.

Figure 1.3: *Robots welding in an automotive factory*

ACCURACY AND RELIABILITY

Humans are emotional beings and can be affected by a variety of internal and external factors. We get tired, we make mistakes and we complain about the work that we need to do. On the other hand, computers are programmed to perform a specific task, in a very specific way and for a set duration. They will perform the task accurately, efficiently and reliably.

In the healthcare industry, technological advancements have improved the accuracy and, therefore, the safety of various medical procedures. For example, laser technology, surgical robots and nano devices are used to increase surgeons' accuracy during operations.

In manufacturing industries, automated machines and robotics have increased the accuracy and reliability of manufactured products.

EFFECT ON TIME AND DISTANCE

The efficiency, accuracy and reliability of computers have changed the way in which we communicate with each other. They have allowed us to do the following:

- Have conversations with friends and family in other countries (VoIP)
- Have business meetings with colleagues in other cities (Skype)
- Send instant random messages to people across the world (email)
- Online banking, which allows customers to pay bills, view account balances, transfer funds from one account to another, pay friends and much more. Financial institutions have also given consumers control over their own security by adding features like the ability to freeze a missing credit card to avoid further charges. Over time, these

controls will only increase as technologies like biometrics and facial recognition keep accounts safe.

- Shopping has become a hassle-free task now and almost anybody can order products online after comparison with other websites. The boom and the resultant competition in the online shopping business are evident. Shopping sites are more interesting because of the huge discounts different companies are offering customers.
- The internet is a very important tool for educators. The internet and its application is user-friendly and make students' life easy. A teacher can use YouTube channels to teach students around the world. Teachers can use a blog in which they can share their career experiences with college graduates. There are various websites for teachers and students to use.

SAVING PAPER
Computers allow users to compile data using spreadsheets, write letters using word processors, send messages using email and complete forms using an online application. Each of these computer-related conveniences reduces the amount of time and effort required to perform these activities, but importantly, they reduce the amount of paper that gets used on a daily basis. This is very important as the trees from which paper is made, play a very important role in reducing the amount of carbon in the air.

GLOBAL COMMUNICATION, INCLUDING SOCIAL NETWORKS AND WEB TOOLS

One of the most-enjoyed attributes of computers is that they can be connected to form a network. This includes connecting the computers in a home, or an office, so that the users can share files. Networks can also span across a distance. The largest network is the internet, which consists of hundreds of thousands of computers across the world.

Computer networks have made it possible for users to use new and exciting ways to upload and share information. Examples of this include the following:

- **Social networks** are specialised computer networks that allow users to have social interactions with each other by sharing their personal information. This includes their likes and dislikes, videos and photos. Some examples of social networks include Facebook, Twitter, Snapchat and Instagram.
- Web tools:
  - **Blogs** are a form of an online diary that allows users to share their daily experiences with others. Examples of blogging websites include "Boing Boing" and "PlayStation Blog".
  - **Wikis** are specialised websites that allow users to share information. This includes all kinds of information; from the plot of a television show, to how photosynthesis works. The most famous example of a wiki is Wikipedia, but there are many other wiki websites dedicated to specific topics.
  - **Vlogs** (or video blogs) are a type of blog where nearly all the content is in video form.

Activity 1.1

d. Which of the following is NOT a reason why we use computers?
  A. Saving time
  B. Lower communication costs
  C. Higher efficiency
  D. Lower accuracy and reliability

e. Which of the following examples of computer uses does NOT help save time?
  A. Using an online site for grocery shopping
  B. Using online banking services
  C. Using a web browser to play flash games
  D. Using Google Maps to find a restaurant's location

2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 A computer	**A.** Time saving
2.2 A benefit of online shopping	**B.** Internet
2.3 Specialised computer networks that allow users to have social interactions with each other	**C.** The user
2.4 The largest network	**D.** Samsung Galaxy S6
2.5 The source of input	**E.** Intel Core 2 Duo
	F. Social networks
	G. Mark Zuckerberg

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if it is false.
  a. Computers have made it possible to talk to people <u>face to face</u> while being in different cities across the world.
  b. The use of automation <u>increases</u> the amount of people needed in the production process.
  c. Humans are <u>more</u> accurate, efficient and reliable than computers.
  d. Computers <u>lower</u> the cost of communication.

4. Answer the following questions:
  a. What is the difference between wikis and blogs? Give an example of each.
  b. Explain what the storage and communication step in the information-processing cycle entails? Give two examples of storage in the information-processing cycle.
  c. How does a computer contribute to saving the environment?
  d. What are the negative implications of computers on society?

1.2 Data, information, knowledge and wisdom

Data is raw, unorganised numbers, signals, or facts. Without first organising or changing it, humans struggle to use data. For example, your school might have data on the names, surnames, addresses, contact details, as well as the results of every class test, assignment, test and exam of all current and past pupils stored on a computer somewhere. While this data is important to store, it could be hundreds, or even thousands of pages long and very difficult to interpret!

Information, in contrast to data, refers to facts and numbers that have been organised so that they are useful to people. For example, if your Mathematics teacher wanted to see how well your current class is performing compared to last year's class, she might ask your school's database to convert its data into averages for the two years. In this way, those thousands of pages of data will be converted into two numbers that can be compared easily. Similarly, the report you receive at the end of each school year takes all the data that the teachers collected throughout the year and turns that data into a single report that you can use to measure your performance.

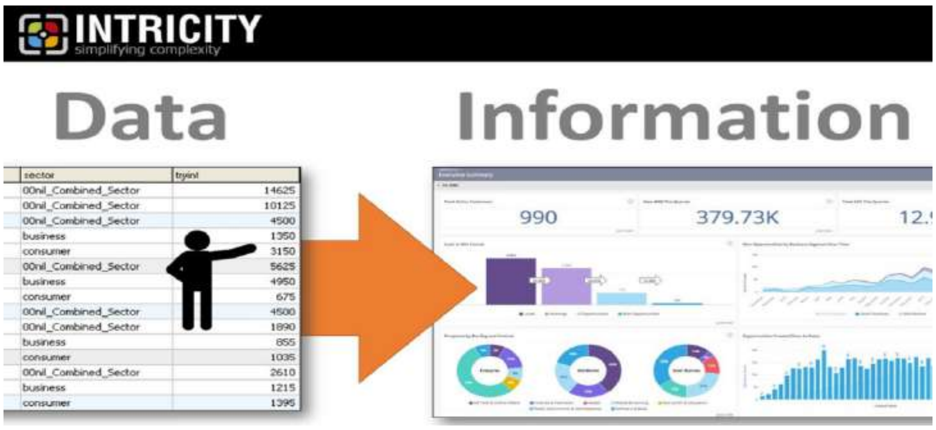

Figure 1.4: *Data is raw facts; information is processed*

Activity 1.2

1. What type of web tool is Wikipedia?
   a. Blog
   b. Wiki
   c. RSS
2. What is the difference between data and information?
3. What software can be used to organise or interpret data in a school environment?
4. Explain to a fellow learner what each step in the DIKW pyramid means and then how it progresses from one step to the next.
5. You just received your CAT test results and comments. The averages of each question and of the test as a whole were given.
   Explain how you would use the data and information to learn from the test and the experience to further deepen your wisdom. For example, how can you and your teacher benefit from the information and how would you learn from your mistakes?

UNIT

1.3 Convergence

Convergence is a term used to describe a situation where multiple technologies are combined to deliver a new and more exciting product.

A smartphone incorporates various technologies that have been combined to deliver a product that can be used for a large variety of tasks. This not only saves you the hassle of having to carry multiple gadgets, but also saves money, as you only have to buy a single product.

To better illustrate this, let's take a look as some of the technologies that have been integrated into the smartphone:

- **Phone:** Like all other cell phones, the smartphone allows you to make and receive phone calls.
- **Camera:** The smartphone contains a camera, which makes it possible to take and view pictures.
- **Video:** You can use the camera on the smartphone to record videos.
- **GPS:** Smartphones come equipped with a global positioning system (GPS), which makes it possible to track your phone and get directions.
- **Music player:** You can use the speaker on your smartphone to listen to music.

Other examples of technological convergence include high-end luxury cars containing computers and video display for parking, smart televisions that allow you to play games and browse the internet, and smart refrigerators from which you can stream music, create a shopping list, and send messages.

Figure 1.5: *Technological convergence*

Activity 1.3

1. Which ONE of the following devices is NOT an example of convergence?
   a. Phablet
   b. Tablet
   c. Standard computer mouse
   d. Smartphone
2. Explain what you understand about technological convergence and give four examples of it.
3. Explain how the convergence of devices benefits the user.
4. What technologies or components have been integrated into the following devices?
   a. Multi-functional printer
   b. A smart device, other than a television or fridge
   c. Virtual reality
   d. Google glass

UNIT

1.4 Social implications: Environmental

There is little doubt that computers have had a massive effect on the world as we know it. Thanks to advances in technology, people all over the world have better access to food and water. They are better educated, more social and wealthier than ever before. However, this is not the only effect that computers have on the world around us.

The data centres used to host websites and the internet use more than 3% of all the electricity generated in the world, and this does not even include the electricity used by personal and work computers. Since most electricity is created by burning coal, computers are a large contributor to the greenhouse gases emitted by humans.

Other potential negative effects include the following:

- **Pollution:** The factories that produce the computers contribute to noise, air and water pollution.
- **E-waste:** This type of waste refers to discarded electronic devices that are thrown away and transported to landfills. Most of these devices contain non-biodegradable materials and heavy metals (lead, cadmium and mercury) that are toxic. The toxins can leak into the ground and contaminate the groundwater.
- **Health hazards:** To extract materials, such as copper, silver and gold from old electronic devices, the old devices are burned. This process releases toxic smoke into the air, which, if inhaled, might cause health problems, such as cancer and kidney disease.

Something to know

The Electronic Product Environmental Assessment Tool (EPEAT) is a system that was developed to help evaluate the impact of computer products on the environment. The method uses three classifications – i.e. gold, silver and bronze. These classifications are based on the materials selection, design for product longevity, reuse and recycling, energy conservation and end-of-life management to evaluate products. Companies, such as Dell and Apple, have started producing products with the aim of meeting EPEAT standards.

THE WAY FORWARD: GREEN COMPUTING

Green computing is the study of designing, manufacturing, using and disposing of hardware, software and networks in a way that reduces their environmental impact. This is normally done by making computers more efficient and making sure that computers are built from biodegradable materials.

Other examples of green computing include the following:

- **Printing:** Using paper and ink in an environmentally friendly way by using recycled paper and printing on both sides of the paper.
- **Saving energy:** Enabling the “Sleep” function on your computer so that it will go into hibernation when your computer is not in use.
- **Disposal:** Properly disposing of and recycling old electronic devices.

Activity 1.4

1. Which one of the following is a potential threat to the environment that is caused by the widespread use of technology?
   a. Increased power consumption
   b. A paperless office
   c. Refilling ink cartridges
   d. Repetitive strain injuries
2. What is meant by the term “e-waste”?
3. What other ways are there to save energy when you use your computer? List at least two.
4 Does the selection of the type of printer also have an impact on green computing? Explain your answer.
5. Do you think that green computing will have any effect on environmental problems? Give reasons for your answer.

REVISION ACTIVITY

PART 1: MULTIPLE CHOICE

1.1 Which of the following devices is NOT an example of a dedicated device? (1)
- **A.** Telephone
- **B.** Barcode scanner
- **C.** DVD player
- **D.** Microphone

1.2 Which of the following computers has the most processing power? (1)
- **A.** Workstation
- **B.** Desktop PC
- **C.** Server
- **D.** Smartphone

1.3 Which of the following computers is portable? (1)
- **A.** Mainframe computer
- **B.** Supercomputer
- **C.** Desktop PC
- **D.** Laptop

1.4 Which step in the information-processing cycle uses the computer's CPU? (1)
- **A.** Input
- **B.** Output
- **C.** Processing
- **D.** Storage

1.5 Which of the following devices uses convergence? (1)
- **A.** Smartphone
- **B.** Television
- **C.** Kettle
- **D.** Digital camera

[5]

PART 2: TRUE OR FALSE

Indicate if the following statements are TRUE or FALSE. Correct the statement if it is false. Change the underlined word(s) to make the statement true.

2.1 A compact computer with a complete system is known as a <u>supercomputer</u>. (1)
2.2 Social networks have helped us to connect people to each other and have, therefore, <u>increased</u> the distance between people. (1)
2.3 <u>Wikis</u> allow us to share our daily experiences with our friends. (1)
2.4 An example of a <u>microcomputer</u> is a laptop. (1)
2.5 People can share their knowledge of the world on <u>social media</u>. (1)

[5]

… continued

REVISION ACTIVITY

… continued

PART 3: MATCHING ITEMS

Choose a term or concept from Column B that matches a description in Column A.

COLUMN A	COLUMN B
3.1 Lowering the brightness of your computer's display screen	**A.** Greenhouse gas
3.2 Discarded electronic devices that are thrown away and transported to landfills	**B.** E-waste **C.** Wikis
3.3 Properly removing and recycling your old electronic devices	**D.** Convergence **E.** Disposal **F.** Blog
3.4 Facts and numbers that have been organised in a way that people understand	**G.** Printing **H.** Knowledge
3.5 A platform where people can share their personal information with others	

[5]

PART 4: SHORT AND MEDIUM QUESTIONS

4.1 How would you use a computer at each stage of the information processing cycle? (4)

4.2 How do you properly dispose of your old electronic devices? (2)

4.3 Describe what convergence is. (2)

4.4 As technology has evolved, computers have changed and, in some cases, improved our lives and ways of living.

- **a.** List five reasons as to why people use computers. (5)
- **b.** List five computing devices you use in your daily life and mention ONE way each of these devices has made your life easier. (10)

[23]

TOTAL: [38]

AT THE END OF THE CHAPTER

NO.	CAN YOU …	YES	NO
1.	Describe the various reasons for using computers?		
2.	Describe the role and use of data, information, knowledge and wisdom as part of information management?		
3.	Explain convergence?		
4.	Explain the social implications of computer technology on the environment?		

TERM 1

HARDWARE

CHAPTER 2

CHAPTER OVERVIEW	
Unit 2.1	Buying the correct hardware
Unit 2.2	Input devices
Unit 2.3	Storage devices
Unit 2.4	Processing devices
Unit 2.5	Output and communication devices
Unit 2.6	Troubleshooting
Unit 2.7	New technologies

By the end of this chapter, you will be able to:

- Evaluate hardware devices.
- Suggest input, output, storage and communication devices, as well as CPU and RAM; including specifying basic specifications in terms of processor, memory and storage for:
  - home users
  - SOHO users
  - mobile users
  - power users
  - disabled users.
- Fix ordinary hardware problems.

INTRODUCTION

Computers and technology have come a very long way since the first computers were used to make advancements in space travel and landing on the moon. They no longer take up entire buildings and cost millions of Rands. Users do not special training either!

SUMMARY OF HARDWARE

The following table summarises the computer components you learned about in Grades 10 and 11, as well as their main functions.

Table 2.1: *Hardware and their purposes*

COMPONENT	TYPES	PURPOSE
INPUT DEVICES		
Keyboard	• Standard keyboard • Laptop keyboard • Gaming keyboard • Virtual keyboard	Enters information, such as letters, words, numbers and symbols into the computer
Pointing device	• Mouse • Touchpad • Touch screen • Trackball • Pen input devices • Joystick	Controls the movement of the cursor on the screen
Scanning and reading devices	• Scanners • Reading devices, such as RFID, QR, OCR and magnetic strip readers	Scans documents, such as photographs and pages of text, and converts them into a digital format – i.e. reading devices to read text, symbols, QR codes and magnetic strips
Video devices	• Video camera • Webcam • Digital camera	Captures media, such as pictures, videos and sound
Audio devices	• Microphones • Voice-recognition device	Communicates with your computer using your voice
Biometric devices	• Fingerprint scanner • Eye and iris scanners • Facial recognition	Measures a person's unique physical characteristics
Point-of-sale (POS) terminals and ATMs		Mainly in the retail and restaurant industry to keep an accurate track of stock and orders; ATMs are used exclusively in the banking industry
PROCESSING DEVICES		
Motherboard		Connects all the components of the computer
CPU		Completes the general processing tasks of the computer
GPU		Completes the graphics processing tasks of the computer
RAM		Very high-speed storage that temporarily stores data that the CPU uses
Read-only memory (ROM)		Non-volatile memory – i.e. when a computer starts up, it uses the information that is stored on the ROM to start up

… *continued*

COMPONENT	TYPES	PURPOSE
STORAGE DEVICES		
Hard-disk drive (HDD)	• Internal hard drive • External hard drive	Slow, long-term storage of data used on the computer
Solid-state drive (SSD)		Fast, long-term storage of data used on the computer
Flash disk		Very small, non-volatile and portable devices that connect to a computer using a USB port
Memory cards		Non-volatile storage devices used mainly for storing digital information
CD, DVD and Blu-ray disks		Portable storage devices to which data files from a computer can be copied, using the correct CD, DVD, or Blu-ray writer
OUTPUT DEVICES		
Display devices	• LCD monitors • Television monitor • Data projector/ DLP device	Displays the images generated by the computer
Printer	• Inkjet printer • Laser printer • Multi-function device • Ink tank printer	Generally used to convert electronic data into a hardcopy
Audio output devices	Headsets and speakers	Audio output devices convert data on a computer into sound
COMMUNICATION DEVICES		
Router		Organises and routes data on and between networks, which may include routing data from a home network to the internet (such as a modem) and connecting many computers to the same network (such as a switch)
Modems		Connects computers to a network and the internet
Switch		Connects many computers on the same internal network

Activity 2.1

1. Write down the correct answer for each of the following questions.
   a. Which one of the following is a technology that is commonly used to connect a variety of different devices to a computer?
      A. OMR B. LCD
      C. USB D. SSD
   b. The ________ retains the data stored on it even if the power goes off.
      A. AM B. HDD
      C. LED D. CPU

… continued

Activity 2.1 *… continued*

c. Which hardware device can create electronic copies of documents by capturing an image?
   A. Printer
   B. Scanner
   C. Stylus
   D. Monitor
d. A(n) ________ is NOT an example of optical storage.
   A. CD
   B. DVD
   C. HDD
   D. Blu-ray disk
e. While editing videos on a mobile device, most of the battery life will be used by the ________.
   A. Microphone
   B. CPU
   C. Screen
   D. Speaker

2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 A storage device used with a card reader	A. Keyboard
2.2 A camera that captures media, such as pictures and videos mostly on a laptop	B. Flash disk
2.3 Enters information, such as letters and numbers into a computer	C. Printer
2.4 Converts electronic data into a hardcopy	D. Fax machine
2.5 A commonly used portable storage device	E. SD card
	F. Webcam
	G. Scanner

3. Correct the underlined word(s) if the statement is FALSE.
   a. A <u>scanner</u> captures media, such as pictures, videos and sound.
   b. A <u>fingerprint</u> scanner scans documents, such as photographs and pages of text, and converts them into a digital format.
   c. A <u>monitor</u> displays or sends data from a computerised device to other users.
   d. <u>Speakers</u> convert data on a computer to sound.
4. Answer the following questions:
   a. What is hardware? Name at least three different types of hardware devices.
   b. What are the two memory components on a motherboard (state their acronyms) and what is the difference between the two?
   c. What is the difference between a modem, router and a switch?

2.1 Buying the correct hardware

When you want to buy a new computer, the most important thing to consider is *how* you will be using the computer. Each computer has different specifications, advantages and limitations, which are linked to each of the components that you will use. A person who uses a computer exclusively for games will have different requirements to a person who needs a computer to browse the internet and do word processing.

To make it easier for you to decide on which components will work for you, we will be taking a look throughout this chapter at some of the things that you have to take into consideration before your next computer purchase.

COMPUTER USERS

People use computers for many different purposes. In this section, we will look at some of the different computer users, by focusing on the following:

- Home users
- SOHO (small office or home office) users
- Mobile users
- Power users
- Disabled users

For each user, you will learn to identify what they are most likely to use the computer for, what components are important for their use and why certain components should be selected.

HOME USERS

As the name suggests, a home user is someone who buys a computer for personal use at home. He or she would usually use the computer to:

- browse the internet;
- post on social networks;
- send emails;
- do word-processing tasks;
- watch online and local videos; and
- listen to music.

However, categorising someone as an "average" home user has become a lot harder than it used to be. A study done in 2017 found that there were 2.2 billion gamers in the world, which is roughly 30% of the world's population. A separate study found that almost 50% of Germans played video games and the average age of gamers in Germany was 35 years old. Many home users also use their computers to do work from home, which makes the distinction between home users and office users, smaller.

Finally, most home-user computers are used by more than one person. This could mean that the children would prefer a gaming computer and the parents would prefer a work or home computer. Due to these complications, a computer that can only do basic tasks, such as browse the internet, or use a word processor, will not work for many households. Instead, households need a flexible computer that can meet the requirements of more than one user.

As a starting point, you can consider a desktop computer with a mid-range CPU and graphics card (for gaming). Most households will also benefit from having a small printer.

If there is only one user in the household and he or she plans to use the computer only for basic tasks, the user should then consider buying a mid-range notebook. Notebooks are more flexible; they can be used in different places in the home and are portable. While budget notebooks (under R5 000) exist, they should be avoided, because they will be slow and uncomfortable to use, even for basic tasks.

SOHO USERS

Small office or home office (SOHO) users use a variety of hardware and devices for their business activities. They may use computers for the following:

- Online research
- Sending emails and business communication
- Using word-processing and spreadsheet applications
- Note-taking
- Printing documents

Although none of these tasks require a very powerful computer, there are four factors that are particularly important to business users:

1. **Mobility** makes it possible to carry the computer around, for example when attending meetings.
2. **Battery life** determines for how long they can use their computers without access to a power source.
3. **Screen** resolution determines how much information they can view on a screen at a time and also affects the quality of the display.
4. **Speed** is an important factor to ensure good productivity.

Based on these requirements, an **ultrabook** is the ideal computer for most business users. Ultrabooks are small and very powerful notebooks with long battery lives. They are easy to carry around and powerful enough to run any business application without slowing it down. Unfortunately, Ultrabooks are very expensive and can cost anywhere between R15 000 and R30 000. As a compromise, most business users would be happy with a mid-range notebook, with emphasis on the factors mentioned above. Even though notebooks have a built-in monitor, mouse and keyboard, purchasing a large, stand-alone monitor with a more comfortable mouse and keyboard is a good investment for many business users and can help to improve productivity.

MOBILE USERS

Mobile users are people who travel a lot. These users require devices that are easy to use and easy to transport. Because of this, mobile users should look at mobile computing devices – such as, tablets, laptops and smartphones.

Mobile users require the following:

- **Mobility**, which allows them to take their devices with them wherever they go.
- **Battery life**, which allows them to use their devices for extended periods without access to a power source.
- At least a **3G connection** to access the internet on the mobile device.

Something to know

Not so long ago, most web browsing took place on desktop and laptop computers. However, this changed drastically with the explosion of smartphones. In a report written by Comscore in 2016, they found that desktop usage is dropping each year with a few percent. At the same time, mobile usage is going up dramatically. The most startling statistic, however, is that one in five millennials (aged 18 to 34) do not use desktops at all. This means that casual browsing is moving away from PCs to mobile devices.

Figure 2.1: *Young millennials using mobile devices*

POWER USERS

Power users need a computer with high processing capacity. They may need the capacity for work-related tasks, such as graphic design, or for personal activities, such as playing games that require high-definition graphics. A power user should buy a computer with a large amount of storage space, enough RAM and a high-end CPU.

Figure 2.2: *Power users often use more than one monitor for their work*

In most cases, power users use computers very similar to those used by gamers, although some power users who travel a lot prefer ultrabooks.

DISABLED USERS

The advances in modern technology have enabled people with disabilities to not only use computers, but to also use computers to make their daily lives easier. The type of computer and its requirements will depend largely on the type of disability and the needs of the user. It is, therefore, important to research the issue before buying a computer for a disabled user.

Something to know

An example of how computers have made people's lives easier is to look at the late physicist, Stephen Hawking. Professor Hawking was a world-renowned, award-winning scientist who suffered from a motor-neuron disease that left him unable to walk and talk. However, thanks to computer technology, he was able to communicate with others and continue to do his ground-breaking research.

BUYING RECOMMENDATIONS

With all of this in mind, we can see that each computer user is unique and has different computer needs. As such, they will require different computer hardware.

The following table lists some recommendations for computer hardware, based on the type of user.

Table 2.2: *Recommendations for computer hardware based on the type of user*

USER	INPUT	OUTPUT	STORAGE	COMMUNICATION DEVICE	CPU	RAM (GB)
Home	• Keyboard • Mouse	• Screen • Speakers • Printer	• Hard drive • CD/DVD drive	• Network card • Router	Entry level CPU	4–8
SOHO	• Keyboard • Mouse • Webcam • Microphone • Scanner	• Screen • Speakers • Printer • Multi function printers	• Hard drive • CD/DVD drive • External hard drive • Flash disk	• Network card • Router • Switch	• Mid-range CPU • Mid-range GPU	4–8
Mobile	• Keyboard • Mouse • Touchpad • Touch screen	• Screen • Speakers	• Hard drive • External hard drive • Flash disk • Memory card	• Wi-Fi card • 3G connection	• Mid-range CPU • Mid-range GPU • (depending on the device)	2–8 (depending on the device)
Power	• Keyboard • Mouse • Joystick • Microphone • Webcam • Scanner	• High-definition screen • Speakers • Headset	• Hard drive • SSD • CD/DVD drive • Flash disk	• Network card • Router	• Top-of-the line CPU • Top-of-the line GPU	8–16

Disabled	• Special design keyboard
	• Special design mouse
	• Joystick
	• Trackball
	• Eye typer
		• Large screen
		• Speakers
		• Printer
		• Specially designed
		  ○ Printer
		  ○ Scanner
			• Hard drive
			• CD/DVD drive
				• Network card
				• Router
				• Specialised software
					• Entry-level to top range CPU (depending on the condition)
					• Entry-level top range GPU (depending on the condition)
						4–16 (depending on the condition)

In the chapters that follow, we will study the different devices in some more detail.

PRODUCTIVITY, EFFICIENCY, ACCURACY AND ACCESSIBILITY ISSUES

Computers are one of the greatest technological advances of the modern era. They bring a variety of advantages, such as helping to improve productivity and efficiency by enabling the completion of difficult and time-consuming tasks in a fraction of the time it would take a human to complete. An example of this can be seen when looking at a factory that produces cars. It could take anything from two months to two years for a person to assemble a complete car. This same task can be completed within 8 hours by simply using a computer.

Computers can be programmed to perform a task in a very specific way. This not only ensures that the task is completed in the quickest way possible, but also that the task is completed as accurately as possible. Another example of computers helping with accuracy can be seen when looking at a database. The computer program allows you to use formulas and functions to do calculations. If you would have to complete these calculations on your own, it would take a significant amount of time and you could run the risk of making mistakes.

The last advantage of computers that we will be looking at is the increase in accessibility. In the past, people with disabilities were limited in the way that they could communicate and complete tasks. Fortunately, computers offer solutions to these problems. Computers not only allow disabled users to communicate with others (through the use of special input devices); they also enable them to perform basic functions, such as driving a car.

Activity 2.2

1. Write down the correct answer for each of the following questions.
   a. Which one of the following is the abbreviation for the largest unit used to measure storage?
      A. Gb
      B. Kb
      C. Mb
      D. Tb
   b. For which type of user is the following list of components intended: Core 2 Duo processor, 4 Gb memory, 500 Gb hard drive and a DVD reader?
      A. Power
      B. SOHO
      C. Home
      D. Disabled
   c. An ultrabook is a preference for which travelling users?
      A. Power
      B. SOHO
      C. Home
      D. Disabled
   d. Mobile users require a device that has which of the following features?
      A. A large amount of processing power
      B. A large amount of storage space
      C. A long battery life
      D. A high-definition display
   e. Which of the following is NOT an example of a computer user?
      A. Entitled
      B. Home
      C. Disabled
      D. Mobile

… *continued*

Activity 2.2

… continued

2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 User type most likely to use an SSD	A. Mobility/3G connection
2.2 A computer with the lowest hardware specifications and is the most affordable in its category	B. Mobile user
2.3 Users most likely to use a computer with the lowest hardware specifications and one that is most affordable in its category	C. Gamer
2.4 The amount of these user types is constantly increasing	D. Power user
2.5 What mobile users require	E. SOHO user
	F. Entry level
	G. Exit level
	H. Home user

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if it is false.
   a. The type of user that uses the most graphic processing power is a <u>home</u> user.
   b. Stephen Hawking was a <u>power</u> user.
   c. <u>Screen resolution</u> determines how much information you can view on a screen at a time.
4. Answer the following questions:
   a. What are the different types of users? Give an example of a hardware component that is most important to each type of user.
   b. What does the acronym SOHO stand for?
   c. Which type of user uses the most processing power?
   d. What is the difference between a SOHO user and a home user?
5. The following specifications appeared in an advertisement for a notebook computer. Study the specifications and answer the questions that follow.

Operating system – Windows 8
Processor – Intel 100 0M Celeron 1.8 GHz
Display – 15.6" (1 366 × 768), 16:9
Video graphics – Integrated Intel HD Graphics 4000
Memory – 4 Gb DDR3, 1 600 MHz memory
Hard disk – 160 Gb, 5 400 rpm
Optical drive – Integrated DVD reader
Sound – Integrated stereo speakers (2 × 1.5 W)
Integrated communications – 802.11n Wi-Fi, 10/100M LAN
Camera – 8 Mp webcam
1 × HDMI
1 × 12-in-1 card reader

   a. Would this computer be suitable for a power user, such as a video-editing professional? Substantiate your answer by giving two reasons to support your answer.
   b. Give two reasons as to why a notebook computer would be more suitable than a desktop computer for learners in a school.

UNIT

2.2 Input devices

To enable you to determine which hardware component suits your needs the best, we will focus on the main input, output, storage and processing devices; with special consideration of their uses, advantages and limitations, as well as other relevant characteristics.

To use your computer or cell phone, you will need an input device. An input device is a piece of hardware that enables a user to enter data into a computer, or interact with a computer. These devices allow you to interact directly with your computer (for example, a keyboard and mouse) and devices where the data is saved or transmitted to your computer (for example, scanners and cameras).

In the following sections, we will look at the keyboard and mouse, their advantages and limitations, as well as the risks associated with the devices. We will also give some advice on how to identify which device is best suited to your needs.

KEYBOARD

The keyboard is the most common and important input device for desktop computers. It consists of various keys that the user presses to give commands or type letters.

ADVANTAGES AND LIMITATIONS OF KEYBOARDS

Most modern computers come with a standard QWERTY keyboard. These keyboards have certain advantages and limitations, as described in the following table.

Table 2.3: *Advantages and limitations of keyboards*

ADVANTAGES	LIMITATIONS
Enables the user to enter information in an easy manner (very little training is needed)	Disabled users might find it difficult to enter information using a keyboard
Serves as a fast method in which to enter data	It is easy to make mistakes while typing in data
Keys can be programmed to serve a specific function	To program the keys of a keyboard for specific functions, it requires a specialised keyboard and knowledge on how to do it
Hotkeys can be used to increase the speed and efficiency of various tasks	The user needs to learn what the hotkeys are and how to use them

ERGONOMIC CONSIDERATIONS

Any person who has worked with a computer for an extended period will know that most computer tasks require a keyboard and a mouse. It is, therefore, important to make sure that your keyboard and mouse are comfortable to use. Here are some factors to take into consideration regarding the ergonomic use of a keyboard:

- Make sure that your keyboard is at the correct height, just above the level of your lap. Your arms should be tilting downward when typing. In many cases, this may mean that you should get an adjustable keyboard tray so that your keyboard can be tilted down and away from you.

Something to know

Never use the kickstands underneath most keyboards.

- Keep your wrists in the neutral (straight) position. If you have to bend your wrist up and down the whole time, it compresses structures in your wrists, and causes pain and injuries, such as carpal tunnel syndrome and tendonitis.

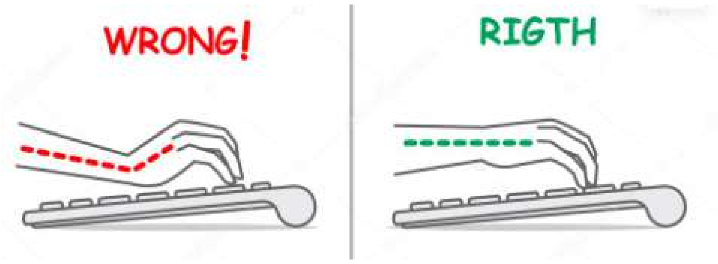

Figure 2.3: *Your wrist must be in a straight position, parallel to the desk*

- Adjust the height of your chair to make sure that your elbows are at a 90-degree angle, or more. Elbows that are bent at less than 90 degrees can cause arm and wrist pain.
- Keep your shoulders relaxed and your elbows at your side. This means that your shoulders should not be raised and your arms should be roughly parallel to the floor.
- Keep a light touch on your keys when typing, as the tendons in your fingers are connected near your elbow. Hitting the keys too hard may cause inflammation of the elbows.
- Align your body to the keyboard, depending on whether you use the letters or the numbers the most. If you use the letters the most, centre the keyboard so that the letter B is about in line with your belly button. However, if you use your **numeric** keypad (or numpad) the most, move the keyboard more to the left.
- Use keyboard shortcuts or macros for common, repetitive tasks to prevent overusing your hands and wrists.
- Buy an ergonomic keyboard if you already suffer from hand, wrist, arm or shoulder pains.

Figure 2.4: *An ergonomic keyboard*

Something to know

Even though there are many types of mice available on the market, a standard two-button optical mouse will be more than sufficient for the average computer user.

MOUSE

The second most common input device to use with a computer is the mouse. A mouse is a pointing device. It allows the user to move the cursor on the computer screen, as well as point, click and select various programs and items. Although the *Oxford Dictionary* uses both "computer mice" and "computer mouses" as the correct plural forms of the term, we will be using the plural form "mice".

ADVANTAGES AND LIMITATIONS OF THE COMPUTER MOUSE

There are very few computer applications that do not require a mouse to work. As such, computers are generally sold with a mouse and a keyboard. It is, therefore, important to take a look at the advantages and limitations of the computer mouse.

Table 2.4: *Advantages and limitations of computer mice*

ADVANTAGES	LIMITATIONS
Easy to use and ideal for desktop and laptop computers	The mouse requires a flat space close to the computer in order to operate
A standard optical mouse is not very expensive	Uneven surfaces might affect the performance of the mouse
They are small and do not take much space	Disabled users might find it difficult to use both a keyboard and mouse
Enables the user to move the cursor and select options faster than using the keyboard	

ERGONOMIC CONSIDERATIONS

There are various health issues that can be caused by using a computer mouse, such as sore wrists, aching elbows and shoulders; all of which cause headaches. It is important to make sure that your mouse is correctly positioned and that you use it in the correct way.

Let's look at the following tips on how to use a mouse ergonomically:

- Computer mice come in different sizes and shapes; therefore, make sure that the mouse fits comfortably in your hand and is easy to use. Rather use a symmetrical shaped mouse than a curved one.
- Do not grip the mouse hard. Hold it loosely.
- Make sure that you are holding your mouse correctly – i.e. rest your hand over the mouse and place your index finger on the left button of the mouse. You can rest your thumb and pinkie on the sides of the mouse.
- Keep your mouse at the correct height and distance from your body, close to the keyboard.
- Make sure that your elbow is bent and close to your body.
- Use your whole forearm, not only your hand, to move the mouse. If you move only your hand, it will put strain on your wrist.
- Keep your wrist straight. This will prevent your wrist from bending in an unnatural position.
- Do not use a wrist rest, as that doubles the pressure inside the carpal tunnel.
- Do not click too hard; rather use a soft touch to manipulate the mouse.
- Adjust the speed at which your mouse moves, the time required between double-clicks, and the size of the cursor if you have problems controlling the mouse.

TOUCH SCREEN

Unlike the keyboard and mouse, the touch screen serves a dual role as both an input and an output device. The touch screen allows the user to use his or her fingers, or a **stylus**, to directly press buttons and select options that appear on the screen.

The most common example of a touch screen can be seen when looking at any of today's smartphones. From Apple to LG, each smartphone uses a touch screen to enable the user to easily navigate and use the various functions of the phone.

ADVANTAGES AND LIMITATIONS OF TOUCH SCREENS

While most modern smartphones rely exclusively on touch screens, there are also many notebooks and tablets that have touch screens.

Table 2.5: *Advantages and limitations of touch screens*

ADVANTAGES	LIMITATIONS
Options can be selected faster than would be possible with a keyboard and mouse	There are a limited number of options available when using a touch screen
The method of inputting data is very user friendly	It is not suited for entering large amounts of data
A touch screen allows for more space as a keyboard and mouse is not required	The screen can get damaged and dirty due to constant touching
The screen interface can be changed by updating the software	Touch screens for computers are more expensive than standard computer screens

TOUCHPAD

The touchpad is a small square or rectangular input device on a laptop. It has the same function as a mouse. You use it by moving your finger across the pad to move the cursor on the screen. Like the standard computer mouse, the touchpad also has two buttons. The left button is used to select objects and the right button is used to bring up a menu. It also has a function that allows the user to scroll up and down a page.

Figure 2.5: *An example of a laptop touchpad*

ADVANTAGES AND LIMITATIONS OF A TOUCHPAD

While touchpads are not an efficient way to move the mouse cursor, they make it possible to use notebooks without having to use any additional devices.

Table 2.6: *Advantages and limitations of touchpads*

ADVANTAGES	LIMITATIONS
Saves space in your laptop bag as you do not need to carry an additional mouse	People with hand or wrist injuries might find it difficult to use the touchpad
Usable when there is no flat surface for a mouse	Harder to control the pointer than with a mouse
Faster to select an option than using a keyboard	More difficult to do actions, such as drag and drop

DIGITAL CAMERAS

A digital camera is used to capture photographs and store them on a digital memory card instead of on film. Once the image is stored, you can transfer it to your computer. You can then manipulate (edit) the image, and print it or upload it to the internet. Some digital cameras can record video images with sound. An example of this is the camera in your smartphone.

ADVANTAGES AND LIMITATIONS OF DIGITAL CAMERAS

Digital cameras have a number of advantages over the older film-based cameras. The biggest advantage is that you can preview the photo you take and delete the ones you do not want to keep.

Table 2.7: *Advantages and limitations of digital cameras*

ADVANTAGES	LIMITATIONS
Easy to use and fast to upload your photos to the computer	Requires the user to be computer literate to make full use of the camera
There is no film that needs to be developed	With some cheaper digital cameras, the resolution might be slightly worse than pictures taken with a traditional camera
Easy to delete pictures that you do not like	Images might need to be compressed to reduce the amount of memory used, which reduces the picture quality
Several hundred images can be stored digitally	Due to being easy to use, the user might take several photos of one object leading to the memory being filled up quickly

RESOLUTION AND IMAGE QUALITY

As you have learned in Grade 11, resolution refers to the amount of detail that a camera can capture. The resolution is measured in **megapixels**. Generally, the higher the resolution, the clearer the photograph or video will be. A lower pixel count means that you can see the sharp edges of each pixel, which makes the image blurry (or pixelated).

Something to know

Digital cameras come packaged with software that allows you to edit photos by adjusting contrast and sharpness. The software can also be used to create and print photo albums. An example of digital camera software includes Digital Photo Professional and ZoomBrowser.

Something to know

It is important to remember that the more expensive digital cameras produce images with a high resolution and excellent quality.

Figure 2.6: *A very blurry and pixelated photo of a dog*

One of the biggest advantages of a digital camera is that it provides the user with a choice of camera resolution and image quality. Although the average user will most likely want to take pictures using the high-resolution setting, you should know that there are some advantages to taking lower-resolution pictures.

Figure 2.7: *A high-resolution photo of an owl*

Images with a lower resolution have a smaller file size and take up less space on a storage device. This saves time when you transfer images. It also saves money, as you can use devices with a smaller **storage capacity**. On the other hand, high-resolution photos have better image quality, allowing the user to crop the image as much as needed with little or no loss of quality.

WEBCAMS

A webcam is a type of digital camera that is connected directly to your computer. It makes it possible for the user to stream live videos to, or through the computer. The camera consists of a lens, image sensor and support electronics. Some webcams include a microphone, allowing you to record sound; others need a separate microphone. Laptops and notebooks have a built-in webcam and in most cases, a built-in microphone as well.

Once a video image is recorded on your computer, you can either save it on the same computer, or upload it to the internet (for example, to YouTube).

Figure 2.8: *An example of a webcam on a laptop*

Something to know

Most webcams come packaged with webcam software, which allows the user to capture images and record videos. It also allows the user to adjust camera sensitivity and enable additional features, such as motion detection. Examples of webcam software include, Acer Crystal Eye Webcam and Logitec Webcam Software.

ADVANTAGES AND LIMITATIONS OF WEBCAMS

Webcams serve a variety of the following important functions in today's society:

- They make it possible for a user to make and upload videos to YouTube and other social networks.
- They make it possible to have a face-to-face conversation with somebody who is far away. This feature is very helpful for business people, as they can have meetings with a group of people who are in different parts of the world. This is known as video conferencing.

Table 2.8: *Advantages and limitations of webcams*

ADVANTAGES	LIMITATIONS
The camera can be left on constantly and activated when the user requires it	Most webcams produce poor-quality videos
Allows people to have face-to-face conversations without the need to travel	The webcam needs to be connected to the computer
Can take pictures without the need of film	Feed when video chatting can be choppy or pixelated
Low cost and high convenience	Webcam-based calls run a greater risk of technological failure due to dropped connections or incompatibility
Compatible with various platforms, such as Windows, Mac, Linux and even some gaming systems, such as Sony PlayStation	Hackers can activate your webcam and watch you without you even knowing it; security information can also be stolen

MICROPHONES

A microphone is an input device that makes it possible for the user to record sound, which is then stored on your computer. A variety of sounds can be recorded; including music, ambient sounds and your own voice.

You can use a microphone with a digital camera or webcam to make videos, to have a face-to-face meeting, or to have a discussion over a chat program, such as Skype.

ADVANTAGES AND LIMITATIONS OF MICROPHONES

While most notebooks and webcams have small built-in microphones, you should buy a specialised microphone if you plan to use your computer to communicate with your voice.

Table 2.9: *Advantages and limitations of microphones*

ADVANTAGES	LIMITATIONS
Makes it possible to communicate with other users over the internet in a fast and efficient manner	Requires an internet connection to communicate with other users
Sound can be manipulated with special software	Sound files can take up a lot of storage space on your computer
Makes it possible to use voice-activation software	Voice-activation software might not always be as accurate as typing on a keyboard

VOICE RECOGNITION

Voice-recognition (or voice-activation) software enables the computer to take verbal commands given by the user, and translate and interpret them. It does this by converting the audio received from the microphone to digital signals that the computer can interpret. The signals are then compared to a database containing words, phrases and actions that should be performed.

Over the last few years, voice recognition has become more common and part of our everyday lives. This can be seen by looking at the artificial assistants that can be found in your smartphone. Apple's *Siri* and Android's *Bixby* both work with voice recognition.

VOICE RECOGNITION FOR MARKETING

In 2017, Burger King in the USA made headlines when it began running a 15-second advertisement on television that specifically targeted Google home speakers and Android phones within earshot. The advertisement consisted of a Burger King employee looking directly into the camera, asking the question "OK Google, what is the Whopper burger?" This prompted nearby virtual assistants to start reading the burger's Wikipedia entry.

However, Google quickly managed to prevent its home speakers to respond to the advertisement by registering the sound clip and then disabling the trigger. Although voices on the television have been found to trigger smart speakers, this advertisement was the first attempt by a company to purposefully hijack users' devices for commercial gain.

Discuss the following questions in class:

1. Have you ever had your device's information "hijacked" for commercial gain? If not, do you know anyone else to whom this has happened?
2. What do you think are the main disadvantages and problems that can be caused by voice-recognition software?
3. What are the advantages? See if you can come up with other examples where voice-recognition software is used to make life easier.

Table 2.10: *Advantages and limitations of voice-recognition software*

ADVANTAGES	LIMITATIONS
Can learn to recognise your own unique speech pattern	Speech must be clear and distinct
Works well for people with physical disabilities	Some versions might require training and setup
Should catch most of the spelling and grammar mistakes	Background noise can influence performance
Can capture speech much faster than the average person can type	Alterations to your voice, such as a cold, might cause problems when using the software
Can be used for specialised voice commands	Programs do not understand the context of language the way humans do, which may lead to misinterpretation

SCANNERS

A scanner is an input device that works a lot like a photocopy machine. However, unlike a photocopy machine, a scanner does not produce a printed copy of the scanned document; it produces a digital copy that you can save on your computer. When the document is in digital format, it can be edited and manipulated before you print it. Examples of scanners include the traditional flatbed scanner, as well as the more modern mouse scanner. The mouse scanner looks almost identical to a traditional computer mouse, but has the advantage that it allows the user to have a scanner at hand when needed. When the scanner is not needed, the mouse scanner can be used as a traditional computer mouse.

Figure 2.9: *An example of a flatbed scanner*

Something to know

Scanners are often packaged with optical character recognition (OCR) software, which converts the text on the scanned document, called a portable document format (PDF), to text that can be used in a word-processing application. This makes it possible for the user to edit a scanned document.

It should be noted that the effectiveness of OCR depends on the quality of the scanned document. Poor-quality scans will make it difficult for the software to function properly.

ADVANTAGES AND LIMITATIONS OF SCANNERS

Scanners are an important part of any office and can also be very useful for private use.

Table 2.11: *Advantages and limitations of scanners*

ADVANTAGES	LIMITATIONS
Images and documents can be stored and printed later	The quality of the scanned document is limited by the resolution of the scanner, as well as the quality of the original document
Makes it possible for the user to edit stored documents	Unable to scan 3D objects
Document quality might be improved by using special editing software	Prone to wear and tear and, therefore, prone to technical difficulties

WIRELESS VERSUS CABLED DEVICES

Over the last 50 years, we have seen many advances in technology. Computers have gone from machines that take up whole rooms, to something the average person can afford and keep on their desk, or even in their pocket. Another advancement was made in the field of wireless technology.

Wireless devices, such as keyboards and mice, are similar to the standard wired versions. However, there are some factors to consider before you decide which type will work best for you.

Something to know

Wired input devices are connected via a wire to the computer, allowing information to travel between the two devices.

Table 2.12: *Advantages and limitations of wired input devices*

ADVANTAGES	LIMITATIONS
Usually cheaper than wireless devices	Wires take up space and might get tangled
Requires no batteries to operate	Wires are not aesthetically pleasing to look at
Little to no delay in communication between the input device and the computer	The device is less portable

Wireless keyboards and mice use a receiver that is plugged into the USB slot on a computer. The input device then communicates with the computer, using a wireless transfer of data.

Table 2.13: *Advantages and limitations of wireless input devices*

ADVANTAGES	LIMITATIONS
Saves space as there are no wires	Battery life might be an issue causing the device to stop working
Aesthetically better than wired input devices	Potential interference of other Wi-Fi signals might cause a decrease in performance
Allows flexibility of use as you can use the device at a distance	There might be some security issues as it is easier to keylog a wireless device

From these two tables, you can see that both wireless and wired devices are similar, with only a few minor differences. The decision of which to buy ultimately comes down to your own choice and circumstance. If you have limited space or need to be able to work from across the room, wireless will be better. If you are on a budget or want to play games, wired will be better.

RISKS ASSOCIATED WITH INPUT DEVICES

The main function of any input device is allowing the user to interact with the computer. Due to this and the fact that most users spend a large amount of time using many of the devices, it is important to note that there are certain risks that you should be aware of. These risks include the following:

- **Keystroke logging:** This is a method used by hackers to record your keystrokes on the computer. The keystrokes can be used to identify personal information, such as bank details and passwords. Keystroke logging, also known as keylogging, is normally done with malicious software (**malware**) that is installed on your computer.
- **Physical injury:** The extensive use of a keyboard, mouse, trackpad or touchpad might lead to physical injuries, such as carpal tunnel syndrome, or repetitive strain injury.
- **Pathogen transmission:** If the input devices attached to a computer are used by a large number of people, it might play a role in the transmission of germs and other pathogens.
- **Privacy issues:** If a webcam and microphone are connected to your computer, a hacker might be able to gain access to them. This will then allow the hacker to gain access to your private images and conversations. As with **keyloggers**, this is usually done by malware that is installed on your computer.

INTEGRATION OF INPUT MODES TO ENHANCE PRODUCTIVITY AND EFFICIENCY

The integration of input devices is a good way to enhance productivity and efficiency. This means that having multiple input methods in a single device can help you to become more productive and efficient with what you do. The mouse scanner is a good example of technology integration, as it combines the traditional computer mouse with scanner technology. The result is a specialised computer mouse that has the ability to scan any size document up to A4, as well as being used as a point-and-click device.

Smartphones use integrated input modes to make users' lives as easy as possible. These methods of input include the following:

- **Touch screen:** The touch screen can be used to play games, type messages and interact with programs.
- **Microphone:** The microphone can be used to talk to people, record messages and for voice recognition.
- **Camera:** The camera can be used to take pictures and videos, and to scan things like QR codes, as well as facial recognition.

On their own, each of these input devices serves an important function. When you combine them into a single device, you get one of the most popular devices the world has ever seen.

Something to know

It is not only the integration of input devices that enhances productivity and efficiency, but also the integration of, for example:

- **Input/output devices:** A four-in-one printer, which you can use to print, copy, fax and scan using just the one hardware device, is a good example of increasing productivity and efficiency with the integration of devices.
- **Custom integrated software and input devices:** Integrating an input device, such as a barcode reader and OCR scanner with an administration system (software), will save time, accuracy and provide comprehensive data during stocktaking or auditing and surveying.

Activity 2.3

1. Write down the correct answer for each of the following questions.
   a. Which of the following components can be used as an integration medium?
      A. Keyboard B. Camera C. Touchpad D. Standard computer mouse
   b. Carpal tunnel syndrome is caused by too much pressure on the wrists. Which of the following devices is most likely to give you CTS?
      A. Monitor B. Printer C. Mouse D. Joystick
   c. Which of the following is a limitation of voice recognition?
      A. Can learn to recognise your own unique speech pattern
      B. Background noise can influence performance
      C. Should catch most spelling and grammar mistakes
      D. Can be used for specialised voice commands
   d. Which of the following is an advantage of the computer mouse?
      A. The mouse requires a flat space close to the computer to operate
      B. Uneven surfaces might affect the performance of the mouse
      C. Disabled users might find it difficult to use both a keyboard and mouse
      D. They are small and do not take much space
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Risk associated with input devices	A. Touch screen
2.2 Smartphone's integration of input	B. Touchpad
2.3 Input device that enables recording of sound	C. Keystroke logging
2.4 Input for *Siri* and *Bixby*	D. Resolution
2.5 Measured in megapixels; the higher, the better the quality	E. Digital camera
	F. Voice recognition
	G. VoIP
	H. Microphone

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if it is false.
   a. It is an <u>advantage</u> for disabled users who might find it difficult to enter information onto a keyboard.
   b. Your wrist must be in a <u>slanted</u> position to use the keyboard correctly.
   c. An optical mouse is very <u>expensive</u>.
   d. The user must be <u>computer literate</u> to make full use of a digital camera.
   e. A <u>touch screen</u> is an input and an output device.
4. Answer the following questions:
   a. What is the difference between a touch screen and a touchpad?
   b. What are two advantages of scanners?
   c. How do integrated systems work?
   d. How do wireless input devices work and what are two advantages of using them?
   e. Which type of user will benefit the least from wireless inputs and explain why?
5. Read through and understand the following scenarios to provide an appropriate solution:
   a. Since your father is technologically impaired, he came to you to explain to him what he must know before buying a webcam for work. List the advantages and limitations of a webcam.
   b. Your parents decide to buy an entry-level computer with internet so that your grandmother can order her clothes and food online, and have them delivered. Explain to your grandmother how to use the keyboard and mouse ergonomically so that she does not suffer from pains. Give her at least seven tips.

UNIT

2.3 Storage devices

In the previous section, we looked at the various input devices that you can use to transfer data to your computer. To ensure that this data is not lost, your computer will need a storage device to store the data, which could either be an internal storage device, such as a standard hard drive, or an external storage device, such as a portable hard drive or flash drive.

To give you a better idea of which storage device is best suited for a specific situation, we will look at some of the most common storage devices, their advantages and limitations, as well as some of the risks associated with them.

Something to know

Storage devices refer to hardware that has a very specific purpose and function, i.e. to store data.

Figure 2.10: *Storage devices from left to right: flash drive, SSD, laptop hard drive and internal hard drive*

Figure 2.11: *An external hard drive*

EVALUATING YOUR STORAGE DEVICE

When deciding to buy a new storage device, there are certain things that you must take into consideration. This includes the following:

- Its **storage capacity**, which determines how much information you can save on the device.
- Its **storage speed**, which determines how quickly new information can be written to the device, or read from the device.

- Its **volatility**, which determines if the device will lose the data when turned off. You do not want a device that will lose all its data in case of a power outage.
- Its **reliability and durability**, which determines how likely the device is to break down. When you store thousands of hours' worth of work or years' worth of photos on a storage device, you do not want it to break unexpectedly.

For internal hard drives, a storage speed of 7 200 revolutions per minute (RPM) is a good starting point, while the capacity will depend on your needs. For most people, 1 Tb should be enough storage space, although you may need more if you plan to store a lot of media files, such as music and videos.

INTERNAL HARD DRIVES

A hard drive is a piece of hardware in a computer in which data is stored and from which you can retrieve data. It is used to store files for your operating system, your software and personal information. Every modern computer comes equipped with an internal hard drive.

ADVANTAGES AND LIMITATIONS OF INTERNAL HARD DRIVES

Modern hard drives have the potential to store up to 12 terabytes (12 Tb) of data. This is enough space to store 3 000 000 compressed songs (such as MP3s), or 17 000 uncompressed CDs. However, these devices are not without fault.

Table 2.14: *Advantages and limitations of internal hard drives*

ADVANTAGES	LIMITATIONS
Large capacity of storage space	Relies on moving parts and is, therefore, prone to wear and tear
Data can be retrieved and saved much faster than from DVDs or CDs	Can easily be damaged if not treated with care
Damaged drives can be easily replaced	Lacks portability, as they are fixed inside the computer
Permanent storage of data	Damage to the drive can cause a loss of data

Something to know

When buying an external hard drive, make sure that you focus on the storage capacity. It is not worth buying a super-fast external drive if it contains a limited amount of storage. For the average user, a 1 Tb external drive will be sufficient. If you only want to back-up some documents and pictures, a flash drive might be a better choice.

EXTERNAL HARD DRIVES

Although most hard drives are located inside the computer, there are some that are used as portable storage devices. These are known as external hard drives. Unlike the internal hard drive, external hard drives are protected by a case that is designed to prevent damage to the drive and is connected to the computer by using a USB cable.

ADVANTAGES AND LIMITATIONS OF EXTERNAL HARD DRIVES

The small size and portability of external hard drives ensure that they can be quickly connected to different computers and are, therefore, ideal for transferring large amounts of data, or backing up data from your internal hard drive.

Table 2.15: *Advantages and limitations of external hard drives*

ADVANTAGES	LIMITATIONS
Large capacity of storage space	Relies on moving parts and is prone to wear and tear
Data can be retrieved and saved much faster than using DVD/CDs	Can easily be damaged if not treated with care
Enables transfer of data between computers	Slightly slower than an internal hard drive
Can be used to back-up the data of your internal drive	Slightly less storage space than an internal hard drive

WHY IT IS IMPORTANT TO BACK-UP YOUR DATA

Tinu is a very busy guy with many social obligations. Due to this, he has been putting off writing his ten-page essay on "Why YouTube is amazing". He knows that he is more than capable of writing this essay in one day; so, he waited until the afternoon before his essay was due. Six hours into writing the essay, there was a minor power surge and Tinu's computer restarted. After the computer started up again, Tinu realised that he forgot to save and most importantly, back-up his work, and that he had to start all over again. Due to this, Tinu rushed his writing and barely finished in time, producing a very poorly written essay.

This example might seem like a minor inconvenience as Tinu still managed to finish his essay in time. Just imagine, however, you have been saving all your photos of family vacations and special moments on your computer, without backing them up on another storage device or medium. A big power surge hits your computer and causes your hard drive to stop working. You have now lost the reminders of all those precious moments. This can be prevented by simply backing up your data.

Backing up data is a simple process where you make a copy of important data on a separate storage device. This device can then be kept at a different location and used to restore any lost data. Traditionally, external hard drives, flash drives, CDs and DVDs are used to back up information.

Discuss the following questions in class:

1. Have you lost important information due to not backing it up?
2. How did that make you feel and what was the impact of it on your life?
3. What could you have done differently?
4. Are you backing important information up? On which device and how do you keep that device safe?

SOLID-STATE DRIVES

The SSD is a storage device (like a hard drive) that does not use the traditional mechanical parts of standard hard drives. Instead, the drive consists of interconnected flash memory chips made of silicon. Due to this, an SSD functions more like a CPU that contains billions of small transistors; each storing one bit of data.

ADVANTAGES AND LIMITATIONS OF SOLID-STATE DRIVES

Solid-state drives, or more commonly referred to as SSDs, are generally many times faster than normal hard drives. However, since a separate transistor is needed for each bit of data stored, SSDs also usually have a much lower storage capacity and a much higher cost per gigabyte. It is, therefore, recommended that you use an SSD to improve the performance of certain programs and not as a primary storage device.

Something to know

For the average user, an SSD of 128 Gb will be more than enough. We recommend that you use a standard mechanical hard drive to store large amounts of data.

Something to know

Flash drives usually have a storage capacity between 4 Gb and 512 Gb, although both bigger and smaller flash drives exist.

Something to know

The flash drive is also referred to as a thumb drive, USB stick, memory stick, or flash stick.

Something to know

CDs have a storage capacity of 700 Mb, DVDs have a capacity of 4.7 Gb and Blu-ray discs have a capacity of 50 Gb.

Table 2.16: *Advantages and limitations of SSDs*

ADVANTAGES	LIMITATIONS
Faster than a standard mechanical hard drive	More expensive than standard mechanical hard drives
Lower power consumption	Limited storage capacity due to high cost per gigabyte
More durable and compact due to the absence of mechanical parts	Shorter life span than standard hard drives or flash drives
No noise, as there are no moving parts	

FLASH DRIVES

A flash drive is a very small (physically) portable storage device. You can connect it to your computer via a USB port. Because it is so small and highly portable, a flash drive is the best way to transfer data quickly and efficiently between two computers. A flash drive is also an ideal storage device for documents and photos.

ADVANTAGES AND LIMITATIONS OF FLASH DRIVES

Flash drives have a very high cost per gigabyte. Due to this, we recommend that you base your purchase on the amount of data that you want to transfer or store.

Table 2.17: *Advantages and limitations of flash drives*

ADVANTAGES	LIMITATIONS
Very compact and portable	Easy to lose, due to its small physical size
More robust than other storage devices	High cost per gigabyte
Does not require a power source to be used	Shorter life span, as the flash memory in the drive can only be used a finite number of times

OPTICAL DISC DRIVES

An optical disc drive is a multi-purpose drive that makes it possible for the user to either read data from optical discs, such as compact discs (CDs), digital optical discs (DVDs) and Blu-ray discs (BDs), or to record data. This, as well as the fact that most movies and music are supplied commercially on these discs, an optical disc drive is one of the most popular storage devices.

ADVANTAGES AND LIMITATIONS OF OPTICAL DISC DRIVES

One of the major advantages of an optical disc drive is that it gives the user an affordable and time-efficient way of producing multiple copies of a disc that contains information. However, due to the limited storage space, this is restricted to small amounts of data.

Table 2.18: *Advantages and limitations of CDs/DVDs*

ADVANTAGES	LIMITATIONS
CDs and DVDs are less expensive than hard drives	Data transfer and access rates are slower than either a hard drive or flash drive
Certain CDs and DVDs can be used multiple times	To reuse the same disc, it requires special rewritable CDs and DVDs, which are more expensive
	Discs can easily be lost or damaged

Something to know

Most optical disc drives are compatible with their predecessors. For example, a DVD drive can read both DVDs and CDs, and a Blu-ray drive can read Blu-ray discs, DVDs and CDs.

CAPACITY AND COST OF THE MOST COMMON STORAGE DEVICES

Throughout this unit, we have taken a basic look at the some of the most commonly used storage devices. We have discussed the advantages and limitations of each device, and have given some indication of which device will be suited for a specific task.

Table 2.19 gives a summary of the capacity and cost of these devices, as well as for which task they are best suited.

Table 2.19: *Cost, capacity and application of storage devices*

STORAGE DEVICE	STORAGE CAPACITY	TASK BEST SUITED
Internal HDD	200 Gb–10 000 Gb	Primary computer storage
External HDD	200 Gb–8 000 Gb	Transferring large amounts of data between computers, or backing up data
SSD	128 Gb–1 024 Gb	Improving the speed of certain important programs, such as your operating system
Flash drive	1 Gb–512 Gb	Transferring smaller amounts of data between computers
Optical drives	CD–0.7 Gb	Distributing small amounts of data to multiple recipients
	DVD–4.7 Gb	Distributing data to multiple recipients

CLOUD STORAGE

One of the advantages of living in the modern world is the luxury of the internet. This global network makes it possible to communicate over great distances. It also provides access to vast amounts of information. The internet has also become one of the best places to store your personal data through cloud storage.

Cloud storage is a type of storage service provided to users that allow them to store information on the internet. The two best known services is Apple's iCloud and Google's Google Drive. Both provide users with a specific amount of storage at no charge (5 Gb for iCloud and 15 Gb for Google Drive), although additional storage space can be purchased at a monthly fee.

Table 2.20: *Advantages and disadvantages of cloud storage*

ADVANTAGES	DISADVANTAGES
Accessibility: It is accessible from anywhere in the world where there is an internet connection.	**Less control over security and privacy:** Cloud computing is vulnerable to attacks. The best example of this was seen on 31 August 2014, when the iCloud accounts of several celebrities were hacked and personal photos leaked to the internet.
Provides disaster recovery: To prevent important information from getting lost due to hardware problems, it can be saved on the cloud from where it can be accessed.	Depending on the service you choose, there can be bandwidth limitations, or vendor lock-ins.
Reliability: If you are with a well-managed service provider, cloud computing is much more reliable than in-house computer infrastructures, and services can be easily transmitted when a server fails.	**Cost:** The cost might increase over the years.
Cost savings: You only pay for what you need. You also save on capital and operational costs, as you do not have to buy expensive hardware, software and storage devices. There is no maintenance.	**Downtime:** Since this service is internet-based, service outages can occur for many reasons.

Activity 2.4

1. Write down the correct answer for each of the following questions.
   a. What current technology is replacing the DVD?
      A. Flash disk B. HDD
      C. Blu-ray D. CD
   b. What is the average capacity of a DVD?
      A. 700 Mb B. 125 Gb
      C. 4.9 Gb D. 4.7 Gb
   c. Which of the following is NOT a storage device?
      A. HDD B. RAM
      C. SSD D. USB stick
   d. Which one is an internal hard drive's average rotational speed?
      A. 3 600 rpm B. 7 200 rps
      C. 7 200 rpm D. 8 400 rps

… *continued*

Activity 2.4

… continued

e. Which of the following is a limitation of an internal hard drive?
   A. Data can be retrieved and saved much faster than from DVDs or CDs
   B. Lacks portability, as they are fixed inside the computer
   C. Permanent storage of data
   D. Large capacity of storage space
f. Which of the following is a limitation of an external hard drive?
   A. Large capacity of storage space
   B. Lacks portability, as they are fixed inside the computer
   C. Slightly slower than an internal hard drive
   D. Slightly more storage space than an internal hard drive
g. Which of the following is an advantage of SSDs?
   A. More durable and compact due to absence of mechanical parts
   B. Longer life span than standard hard drives or flash drives
   C. Less expensive than standard mechanical hard drives
   D. Increased storage capacity due to high cost per gigabyte
h. Which of the following is an advantage of flash drives?
   A. Low cost per gigabyte
   B. Does not require a power source to be used
   C. Difficult to lose due to physical size
   D. Longer life span, as the flash memory in the drive can be used an infinite number of times
i. Which of the following is an advantage of optical drives?
   A. Data transfer and access rate is higher than either hard drive or flash drive
   B. To reuse the same disc requires special rewritable CDs and DVDs, which are less expensive
   C. Discs are robust and cannot easily be lost or damaged
   D. Certain CDs and DVDs can be used multiple times
j. Which of the following is NOT an advantage of cloud storage?
   A. Accessibility
   B. Reliability
   C. Control
   D. Disaster recovery

2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Primary computer storage	A. Storage device
2.2 Device most suited for backing up data	B. Optical discs
2.3 Storage medium that can easily be scratched or damaged	C. Internal HDD
2.4 Storage accessible anywhere in the world as long as there is internet	D. Cloud storage
2.5 Personally transferring a smaller amount of data between computers	E. External HDD
	F. SDD
	G. Flash disk

… continued

Activity 2.4 *… continued*

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if it is false.
   a. Another name for a flash disk is a <u>thumb drive</u>.
   b. A flash drive is the <u>least</u> robust storage device.
   c. An internal <u>hard drive</u> has the highest storage capacity.
   d. A hard drive has a <u>shorter</u> life span than an SSD.
   e. Cloud storage is <u>for free</u> for a specific amount.
4. Answer the following questions:
   a. What is the difference between an internal and an external hard drive?
   b. What is the difference between an HDD and an SSD?
   c. How would you explain cloud storage to a Grade 10 CAT learner?
   d. What five points must you take into consideration when choosing a storage device? Explain these points briefly.
5. Read through and understand the following scenarios to provide an appropriate solution:
   a. Your parents gave you R1 500 to buy a storage device for your computer. Explain the factors involved in your decision to either buy a hard drive or an SSD. State your choice at the end.
   b. Your uncle wants to open up a store that sells music CDs and DVDs, as well as clean CDs. Give him advice on the relevancy and future of CDs and DVDs. Give your opinion on whether or not it will be wise to open such a store and state why.

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2.4 Processing devices

The data received from an input device must be modified and changed before it can be sent to the computer's output device. This is done by the computer's processing devices through the implementation of certain instructions and calculations.

In this section, we will take a brief look at the most common processing devices. This includes the CPU, GPU and RAM. We will also provide some guidelines on how to determine which processing device is best suited for you.

CENTRAL PROCESSING UNIT

The CPU is one of the most important parts of any computer. It is responsible for the following tasks:

- Receiving and carrying out the computer's instructions
- Allocating more complicated tasks to other chips that will better handle the task

The CPU functions by using billions of microscopic transistors that can each be switched on or off individually. As the name suggests, it is the core or centre of any computer.

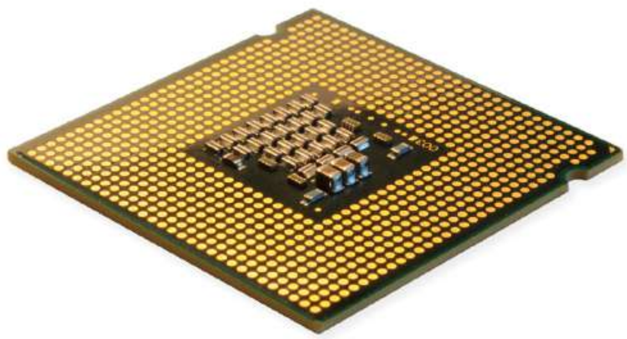

Figure 2.12: *A CPU*

EVALUATING A CPU

When trying to decide which CPU is best suited for your needs, you should consider the following:

- **Gigahertz of the processor:** Gigahertz refers to the CPU frequency and is an indication of the processor's speed. As a general guideline; the higher the frequency, the better the CPU.
- **Number of cores:** As with the processor speed, generally more cores are better.

However, as with many of the other factors, this is not a fool-proof method, as computers with the same number of cores and the same gigahertz can operate at very different speeds. The best advice is, therefore, to research CPUs on the internet and find the best CPU in your price range.

Table 2.21 and Table 2.22 give a summary of some of the most commonly used CPU models.

Table 2.21: *Celeron and Core processors*

MODEL	TARGET AUDIENCE	COMMENTS
Celeron	Entry level	The slowest of the models
Core i3	Mid range	Good starting point for most people
Core i5	Suitable for most uses	Very fast CPU that is suitable for most users
Core i7	Gamers and power users	Faster and more expensive
Core i9	Power users who need cutting edge technology	Fastest and most expensive currently available

Table 2.22: *Ryzen processors*

MODEL	TARGET AUDIENCE	COMMENTS
Ryzen 3	Entry level	The slowest of the Ryzen models
Ryzen 5	Mid range	Good starting point for most people
Ryzen 7	Suitable for most uses	Very fast CPU that is suitable for most users
RyzenThreadripper	Top of the line	Faster and more expensive than any other model

GRAPHICS PROCESSING UNIT

The GPU is a specialised processing unit responsible for display functions. Like the CPU, the GPU is responsible for making calculations and following instructions. However, unlike the CPU, the GPU's instructions are limited to the calculations needed to render and display images on the screen.

Figure 2.13: *A GPU on a PCB board*

Something to know

Take note that even though the Core i7 is faster than the i5, it is also a lot more expensive. For the average user, this difference in speed will not be noticeable and as such, the i5 is much better value for your money.

INTEL VS AMD

Which CPU platform should you buy into right now?

https://www.youtube.com/watch?v=0rBJtC0L2GQ

Something to know

In modern 3D games, thousands of objects can appear on the screen at the same time. Each object is a complex shape with its characteristics determined by using mathematical formulas. For example, to ensure that a 3D world looks realistic, the GPU may calculate thousands of beams of light that bounce around the world to create the light and shadows of the scene. This requires billions of calculations per second, and, as a result, GPUs often have more processing power than CPUs.

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EVALUATING A GPU

When you try to decide which GPU is best suited to your needs, ask yourself the question: "What am I going to use my computer for?" If your answer is "general use", such as email, word processing and photos, then you might not even need to purchase a GPU. Most modern computers come equipped with a fairly powerful onboard GPU that is more than capable of performing the actions needed by the average computer user. However, if you plan to use your computer for graphically intensive tasks, such as graphic design or gaming, you will definitely have to consider an advanced GPU.

Unfortunately, buying an advanced GPU is not as easy as it sounds. There are various manufacturers that have a variety of products with different specifications. To find the correct product for your needs, you have to look at the manufacturer's model numbers and benchmarks. Looking at the model number will provide you with an indication of which generation the card is, while the benchmarks will show you how powerful it is.

Table 2.23: *An example of GPUs*

MODEL	TARGET AUDIENCE	COMMENTS
X30	Entry level	The slowest of the models
X50	Mid range	Good starting point for most users
X60	Suitable for most users	Very fast GPU that is suitable for most users
X70	High end	Very fast GPU that will allow you to run the most complicated 3D applications
X80	Top of the line	Fastest and more expensive of the models

With these model numbers, the X is replaced by a number that refers to the generation of the graphics card. For example, the GTX 980 was released in March 2015; while the GTX 1080 was released one year later in May 2016. This model was followed by the RTX 2080 that was released in September 2018.

While these model numbers and the patterns behind them may change, spending a few minutes to understand what the different numbers mean, will allow you to pick a graphics card suited for you at an appropriate price.

RANDOM ACCESS MEMORY

RAM is a piece of hardware (chip) responsible for temporarily storing data. It also enables the computer to work with a vast amount of information at the same time. It does this by acting as a helper to the GPU and CPU, storing the information that is currently being used, and loading the data that they may want to use next.

HOW DO GRAPHICS CARDS WORK?

https://www.youtube.com/watch?v=ZfnPFNnXqC0

Something to know

The GPU is also referred to as the display adapter, graphics card, video adapter, video card, video controller or even gaming card.

Something to know

RAM can be as much as 90 times faster than the fastest storage devices.

Figure 2.14: *RAM*

Something to know

Take note that your motherboard will determine which type of RAM you will need to buy. The most common varieties include the following:

- **DDR2 SDRAM:** This RAM is normally found in older computers. It was succeeded in 2007 by the DDR3 SDRAM.
- **DDR3 SDRAM:** This RAM is normally found in computers made after 2007. It was succeeded in 2014 by the DDR4 SDRAM.
- **DDR4 SDRAM**: This RAM is found in most modern computers.

The best advice is to do some research on the internet to find out which RAM will work best with your motherboard.

EVALUATING RAM

To be able to both store data and load new data fast enough to let the GPU and CPU operate without any interruptions, RAM needs to me much faster than any other storage device. Having a processor that can complete four billion instructions per second is not useful if it cannot be provided with four billion instructions each second. It is, therefore, very important that your computer is equipped with enough memory to allow the GPU and CPU to operate at their full potential.

Determining how much RAM is needed will depend on the following two main factors:

1. Which operating system are you using?
2. What are you planning to do with your computer?

Table 2.24: *Recommendations with regards to RAM*

TYPE OF USER	RECOMMENDATION	ACTIVITIES
Casual user	4 Gb–8 Gb	Internet browsing, email, watching movies and listening to music
Frequent user	8 Gb–16 Gb	Simple graphic programs and flash games
Power user	16 Gb	Photo editing, video editing and gaming
Professional user	32 Gb	High-performance gaming and graphic design

Activity 2.5

1. Write down the correct answer for each of the following questions.
   a. How much faster can RAM be than the fastest storage device?
      A. 90 times B. 80 times
      C. 70 times D. 100 times
   b. Most modern computers use this type of RAM.
      A. DDR2 SDRAM B. DDR3 SDRAM
      C. DDR4 SDRAM D. DDR5 SDRAM

… *continued*

Activity 2.5

… continued

c. Which amount of RAM is recommended for a professional user?
   A. 32 Gb   B. 16 Gb
   C. 8 Gb   D. 4 Gb
d. Which of following can be used to decide on a CPU?
   A. The amount of heat produced
   B. Number of pins
   C. Number of cores
   D. The amount of data that can be stored
e. Which of the following is responsible for temporarily storing data?
   A. RAM   B. GPU
   C. CPU   D. HDD

2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Other name for a GPU	A. Graphics card
2.2 Volatile temporary data storage	B. CPU
2.3 Processing unit for displaying	C. GPU
2.4 Functions by using billions of microscopic transistors for all calculations	D. RAM
2.5 User uses 8–16 Gb RAM for simple graphic programs and flash games	E. ROM
	F. DDR3
	G. Power user
	H. Frequent user

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if it is false.
   a. A home computer or <u>entry-level user</u> does not need a GPU.
   b. <u>All types</u> of RAM fit on any motherboard.
   c. A <u>GPU</u> is the core of any computer.
   d. A GPU is located on the <u>motherboard</u>.
   e. The more cores a <u>CPU</u> has, the more prone it is to gain heat and slow down the processing speed.

4. Answer the following questions:
   a. For what is the CPU responsible?
   b. What must you consider when evaluating a CPU?
   c. What is the difference between a CPU and a GPU?
   d. What is RAM and what does it do?
   e. What must you consider when evaluating a GPU?

2.5 Output and communication devices

In the previous unit, we looked at the devices that feed data into your computer, the hardware that is responsible for storing the data and the hardware that processes the data. We will now take a closer look at some of the most common devices that are used to present the processed data to the user, or to other computers and networks.

OUTPUT DEVICES

An output device is any device that takes the data that has been stored on a computer and makes it available to the user. This can be done visually (such as, a computer screen or printer), or auditory (such as, speakers).

COMPUTER SCREEN

The computer screen or monitor is the most important output device of any computer. It makes it possible for the user to visually interact with data and programs in a quick and easy manner. This is possible because the computer software is built around a visual representation of data; whether it is a page of text, video on the internet, or 3D elements in a game.

EVALUATING A COMPUTER SCREEN

When looking to buy a new computer screen, there are three main things to look at:

1. **Pixels:** A good number of pixels is 1 920 pixels across the width of the monitor and 1 080 pixels across the height of the monitor. This is normally represented by the resolution of the screen as 1 920 × 1 080, or full HD – i.e. the higher the number of pixels, the better the screen.
2. **Screen size:** The size of a computer screen is measured diagonally from the bottom of the screen to the top right. The measurement is made in inches and one inch is roughly 2.5 cm long. When looking at the size of the computer screen, you should take into consideration how big your desk is. A big screen can cause eye fatigue and even damage to your sight if not placed at the correct distance from your face. Any screen bigger than a 27-inch monitor will be too big for a normal-sized desk.
3. **Refresh rate:** This refers to the amount of time it takes for the screen to be updated with the newest information. It is normally represented as the number of times the screen updates in one second. Most modern computer screens have a refresh rate of 60 Hz. This means that the screen is updated with new image information 60 times in one second. This should be more than sufficient for most users. However, there are advanced screens available that have a refresh rate of up to 144 Hz. These screens, like those with a higher number of pixels, will be much more expensive.

Something to know

Modern monitors can have up to 3 840 × 2 160 pixels (called "4K" resolution). This, however, can be an expensive purchase.

VIRTUAL REALITY

Virtual reality or sometimes abbreviated as VR, is one of the most exciting and interesting advances of technology. It is a type of output device that uses a specially designed headset to fully immerse the user in a high-quality 3D virtual world using both sight and sound.

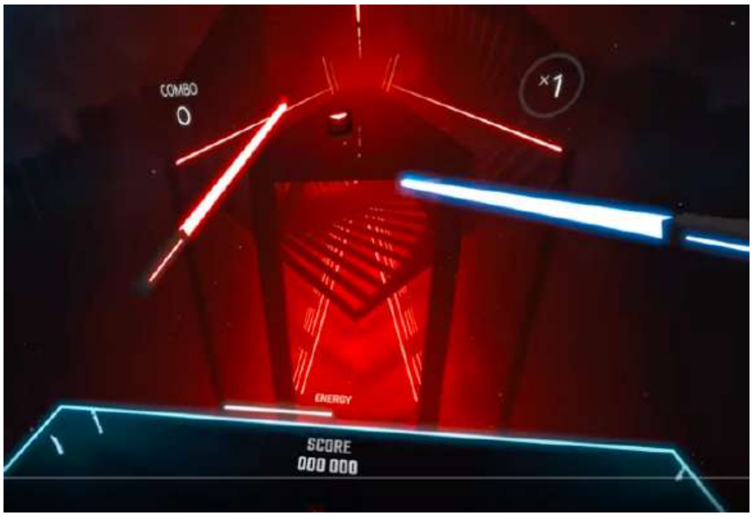
What sets virtual reality apart from other 3D viewing devices is that it is not limited to just normal 3D images. The virtual reality headset adjusts your vision, depending on where you look in real life, changing the picture you see as you look at different things. This gives the device a much more realistic feeling.

… continued

VIRTUAL REALITY

Discuss the following questions in class:

1. Who in your group has experienced virtual reality? How does it feel?



2. Would you like to have virtual reality available to you? Why or why not?
3. Apart from gaming, what else can virtual reality be used for? Do some research on the internet and find at least five different applications.

VIRTUAL REALITY

Watch the following video:

Beat saber
https://www.youtube.com/watch?v=9Rqgu2d3QG8

SPEAKERS

Speakers are one of the most popular output devices. They play back the sound on your computer. This makes it possible to listen to music, speak to friends over Skype, or watch movies.

EVALUATING COMPUTER SPEAKERS

Most computers have their own built-in speakers, which are fine for hearing the normal computer sounds. However, if you want to listen to music, movies or games, you should get additional speakers. The easiest option is to buy a set of computer speakers that you can just plug into the audio jack on the computer.

Let's take a look at some of the following tips of what to look for when you buy speakers:

- **Speakers with or without a subwoofer (2.0 or 2.1 speakers):** A subwoofer is a loudspeaker designed to reproduce low-frequency sound (for example, bass sounds). Table 2.25 compares speaker systems with and without subwoofers.

Table 2.25: *Comparison of a 2.0 and 2.1 speaker system*

2.0 (NO SUBWOOFER)	2.1 (WITH SUBWOOFER)
Has two channels (left and right) and no subwoofer	Has two channels (left and right) with a subwoofer
The amplifiers housed inside one of the speakers	Uses smaller left- and right-hand speakers for higher frequencies and a subwoofer that produces the lower frequencies
Is smaller and takes up less space	Can reproduce lower frequencies and a more powerful bass

- **Specifications and sound quality:** The best way to evaluate the quality of sound produced by the speakers is to go and listen to them. If that is not possible, do your research on the internet and read the reviews of sources you trust. A good speaker system usually provides a good balance between the treble (high), mid-range and bass (low) frequencies; producing a full, rich sound while keeping the detail.
- **Input:** Look for a system that gives you more than one audio-input jack so that you can also plug in your iPod, smartphone, or other audio devices.
- **Control:** Although the systems that provide no control are the cheapest, it is better to get a system with at least a volume control. Some systems let you adjust the bass and treble levels as well. When you buy a system with more controls, make sure that they are easy to reach.
- **Appearance:** Make sure that you like the way the speakers look, as you are going to see them whenever you are sitting at your computer.
- **Price:** It is mostly true that the more you pay for speakers, the better they are. However, make sure that you shop around, as speakers are some of the most discounted computer accessories around.

PRINTERS

A printer is a device that makes it possible to transfer text and graphic output data from your computer to a piece of paper. You can print pages of text, illustrations, diagrams and photos.

WHICH PRINTER IS BEST FOR THE TASK?

When you buy a printer, it is very important to know what you will be using the printer for, since different printers are good at different things. If you are looking at printing only black and white documents, you do not need to purchase a colour printer, as it will cost more to operate. Other things to keep in mind include the following:

- **Speed:** Budget printers should be able to print about three to six colour pages per minute.
- **Paper type and size:** Make sure to select a printer based on the type of documents you have to print.
- **Printer resolution:** This is normally measured in dots per inch (dpi), and affects the sharpness of the text and images on the paper. A resolution of 600 dpi should be sufficient for the average user.
- **Costs to print a page:** Colour printers are more expensive to operate and will cost more per page. This is also true for printers with a higher resolution.
- **Memory:** All printers come with a limited amount of memory. Having more memory increases the speed at which printing takes place. This will only be noticeable if you are planning on printing pages with large images or tables on a regular basis.
- **Wireless capabilities:** Most modern printers come with Wi-Fi capabilities. This makes it possible for the user to print documents without the need to be physically connected to the printer.
- **Ink cartridges:** Make sure that ink cartridges are available for when you have to replace them.

COMMUNICATION DEVICES

A **communication device** is a type of device that connects a computer to other computers in a network. Examples of communication devices include internal equipment, such as a network or Wi-Fi card, as well as external equipment, such as a router.

Built-in communication devices are usually designed to communicate with a very specific type of network. For example, a computer network card will either allow you to connect using a physical ethernet cable, to a Wi-Fi network, or to the internet. In contrast, the purpose of a networking device, such as a router, is to connect many different types of networks, so that it may connect devices using ethernet cables, Wi-Fi, DSL and 3G on the same network.

Activity 2.6

1. Write down the correct answer for each of the following questions.
   a. What is the refresh rate of a computer screen measured in?
      A. Frames per minute B. Hz
      C. rpm D. m/s
   b. What is the most limiting factor when buying an output device?
      A. Cost B. Performance
      C. Quality D. Durability
   c. Which of the following is NOT an output device?
      A. Speakers B. Virtual-reality glasses
      C. Scanner D. Touch screen
   d. Which of the following is an important factor when buying a printer?
      A. Power usage B. Size
      C. Memory D. Lens quality
   e. Which of the following is NOT a factor when buying speakers?
      A. Input B. Control
      C. Appearance D. Storage
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Transfers text and graphics to paper	A. Subwoofer
2.2 A communication device	B. Printer
2.3 The output device is a headset	C. Flash card
2.4 Speaker producing a low-frequency sound	D. Treble speaker
2.5 Printer and scanner integrated into one device	E. Virtual reality
	F. Scanner
	G. Multi-functional printer
	H. Wi-Fi card

… continued

Activity 2.6

… continued

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if it is false.
   a. The bigger the screen size, the <u>better</u> it is for your eyes.
   b. When a screen is described as 1 920 × 1 080, it means that the screen has 1 920 pixels along the <u>height</u> of the monitor.
   c. It is possible to listen to music on your computer without a <u>dedicated sound card</u>.
   d. The price of any output device is <u>always</u> a factor when evaluating the output device.
   e. Having more memory in a printer increases the <u>printing speed</u>.
4. Answer the following questions:
   a. Define an output device and give three examples of it.
   b. How does a speaker system with a subwoofer compare to a speaker without one?
   c. What are the three most important aspects you must look at when buying a monitor?
   d. What five aspects must you look at when buying a printer?
   e. Virtual reality equipment is becoming more affordable. In what way can virtual reality be used? Name two pieces of equipment that one would need for virtual reality.

UNIT

2.6 Troubleshooting hardware devices

As with most things in life, computer equipment is not without fault. As hardware gets older, wear and tear sets in, leading to a variety of potential issues. To make your life easier (and hopefully save some money), we will be taking a look at some of the most common issues that you might encounter, as well as how to resolve them.

In this unit, we will look at some of the most common problems that people experience with their computers. We will also look at ways in which to solve these problems.

ERRATIC MOUSE MOVEMENT

Erratic mouse movement is one of the most common issues that might occur when you use a mouse. This happens when your mouse pointer starts acting erratically, jumping across the screen and does not move properly.

Let's take a look at some of the possible causes and solutions to fix erratic mouse movements:

- **The mouse is dirty:** The performance of an optical mouse might be affected by dirt inside the mouse. Removing the dirt by cleaning the mouse should fix the issue. Be sure to look up how to clean the mouse properly before you do it.
- **Bad surface:** An irregular or shiny surface might interfere with the optical laser that is used by your mouse. This can be fixed by getting a mousepad or book to work on.
- **Wireless mouse:** If you are using a wireless mouse, the issue might be caused by low battery power. Replacing the batteries should fix the issue.

SCANNING

Scanning a document does not always produce the expected result. Let's take a look at some of the most common issues that you might encounter, as well as their possible solutions:

- **A skew or cut-off scanned document:** This is caused by a document that is not properly aligned. To fix it, reposition the document on the scanner.
- **Poor image quality:** This can be due to two reasons:
  - Caused by a dirty scanner surface: Cleaning the surface should resolve the issue.
  - Caused by the inability of the OCR software to read the text: This can be fixed by making sure that the document you want to scan is clear and readable.
- **Connection errors:** This is most likely caused due to a poorly attached connection. Make sure to properly connect the scanner to fix the issue.

DISK ERRORS (DEFRAGGING)

Hard drives are prone to wear and tear, which can lead to a variety of problems, such as disk errors caused by disk **fragmentation**. Therefore, it is very important to regularly check the condition of the hard drives to make sure that they do not fail. In order to do so, you can do the following.

Guided Activity 2.1

1. Open your computer's *Start* menu.
2. Enter "command prompt". Then, click the *Command Prompt* option. You should now see a window open.
3. Type in the following command: "wmic diskdrive get status". This command will look at the key health indicators of your hard drives and see if there are any problems.
4. Press *Enter* to see the status of your hard drive.

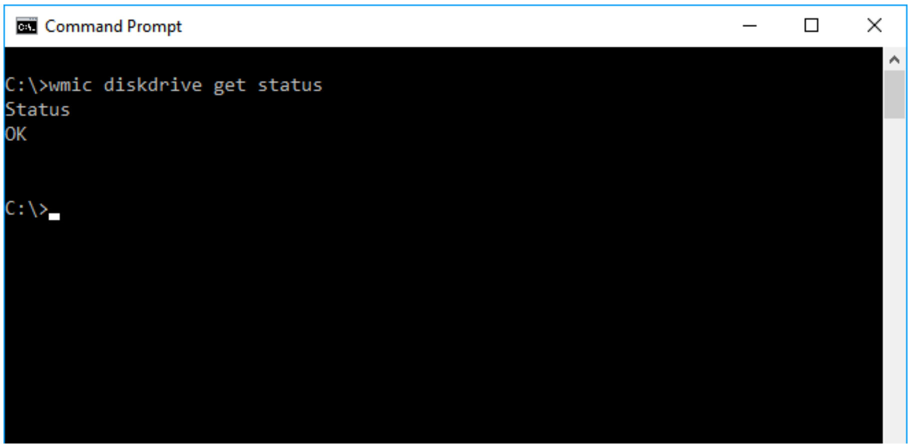


Figure 2.15: *Checking the status or condition of your hard drive*

5. If you receive the "Status OK" message, your hard drives are in a good condition. If not, you should immediately back up the important files from your hard drive and try to replace it as soon as possible.

Another tool that you can use to maintain your hard disks is the *Optimize Drives* tool (previously called the *Disk Defragmenter*). This tool will help you to reorganise the way in which the data is stored on your hard drives (without changing any files), so that your drives can run smoothly.

Guided Activity 2.2

1. Open your computer's *Start* menu.
2. Enter the words "optimize drives".
3. Click the *Defragment and Optimize Drives* application. The *Optimize Drives* window should open.

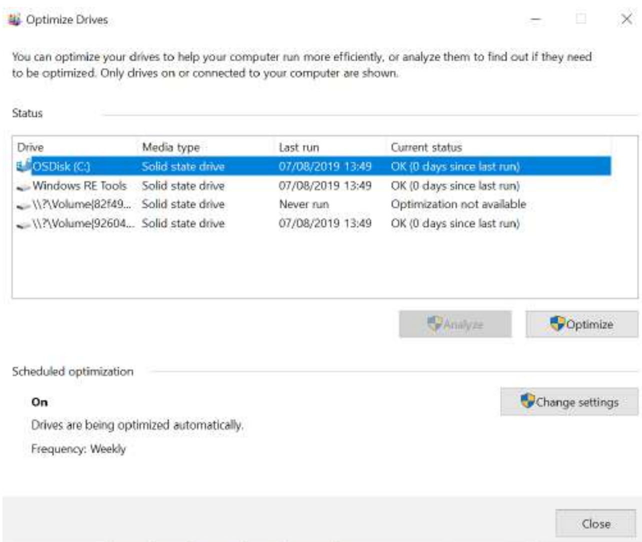


Figure 2.16: *The Optimise Drives window*

… *continued*

Guided Activity 2.2

… continued

4. Click the *Optimise* button to begin optimising your hard drive.
5. At the bottom of the *Optimise Drives* window, the *Scheduled optimisation* should be enabled. If not, you can enable it by using the *Change settings* button.

Something to know

SSDs should never be optimised or defragmented. By default, Windows 10 will not attempt to optimise these drives.

RESOLUTION

As previously discussed, resolution refers to the number of pixels on the computer screen and plays an important role in determining image quality. We will now look at some of the most common resolution-related issues that you might encounter, as well as how to fix them.

Black screen:

- The screen is not plugged into the computer. Plugging in the screen should fix this issue.
- The setting of the screen resolution is too high. Operating systems, such as Windows, have a built-in feature that allows you to return to your previous resolution 15 seconds after you selected a new one. Just make sure that you do not click on *Keep these changes*.
- The images and text on the screen look very big. This is most likely because the resolution is very low and changing the resolution should fix the problem. This issue is sometimes caused when updating your computer's graphics drivers.

Guided Activity 2.3

1. Open your computer's *Start* menu.
2. Enter "display settings" and click the *Change Display Setting* option. You should now see a new window open. (The *Display Setting* window can also be opened by right clicking on your desktop and selecting the *Display Setting* option from the menu.)

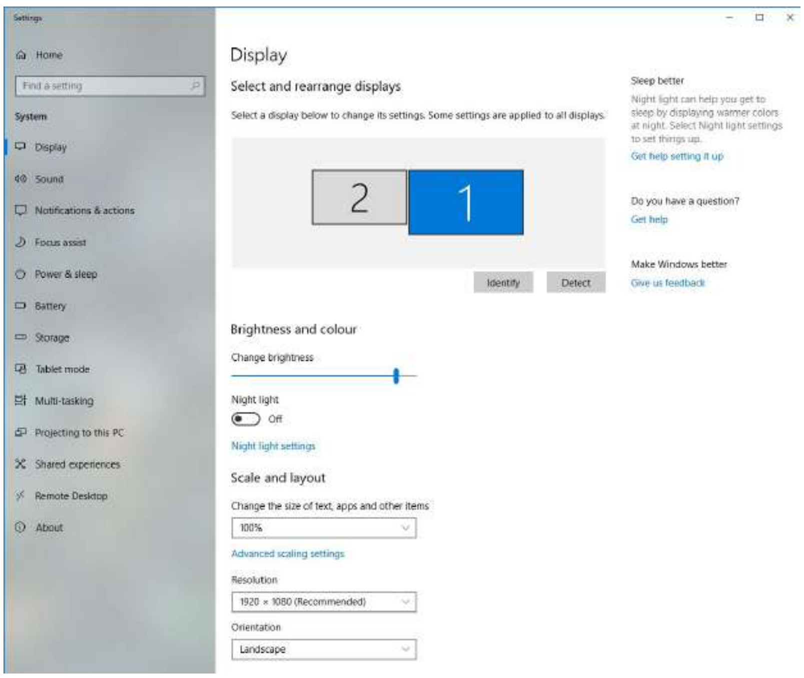


Figure 2.17: *The Display Settings window*

… continued

Guided Activity 2.3

… continued

3. Scroll down to the resolution section and select an appropriate resolution from the drop-down menu. (There should be a recommended resolution that you can use.)
4. Click to keep these changes and close the *Display Settings* window.
5. If you encounter a black screen when changing the resolution, do not worry as Windows 10 has a built-in feature that allows you to return to your previous resolution 15 seconds after you have selected a new one.

NON-RESPONDING PROGRAMS, MOUSE AND KEYBOARD

One of the most common problems that you might encounter when working on a computer, is that the program you are using, stops responding. This might be caused by a variety of factors, such as faulty programs, your computer running out of resources, or faulty hardware. The best way to resolve this issue is to close the non-responding program and then to re-open it. However, it is difficult to close programs that are not responding, as they do not respond to clicking, or selecting the *Close Program* option.

Guided Activity 2.4

1. Open your computer's *Start* menu.
2. Type in "task manager" and click the *Task Manager* option. You should now see a new window open. (You can also open *Task Manager* by pressing the *Ctrl*, *Shift* and *Esc* keys at the same time, or you can right click on the taskbar.)
3. Select the program that is not responding. There will be a list of all the programs that are currently running on your computer. Scroll down and select the non-responding program.
4. Click the *End Task* button. This should close the non-responding program.
5. Once the non-responding program has been closed, re-open it again.

PRINTING PROBLEMS

Printing problems is one of those things that every person will suffer through at least one point in their lives. Let's take a look at the some of the most common printing problems and their possible solutions:

- **Printing takes too long:** This can be caused by a high-resolution setting, memory issues, or printer drivers.
- **High-resolution images:** High-resolution images require more data from your computer, which can slow the printing process down. This can be fixed by selecting the standard or normal mode. This also works to resolve memory issues.
- **Printer software:** Make sure that your printer software is updated. If not, updating the software could fix the issue.
- **Paper jams:** This is caused by a piece of paper becoming stuck inside the printer. To fix the problem, start by inspecting the paper path and removing any stuck papers.
- **Bad printing quality:** Check that your printer has enough ink or toner and that it contains the correct paper.
- **Printer does not print:** This might be caused by a bad connection, the printer running out of ink or toner, or the printer driver not being installed on your computer.

LACK OF FREE SPACE ON STORAGE MEDIUM

The storage on your computer is limited and as such, you might come across a message telling you that you do not have enough hard-drive space to copy, or save a file. This issue can be resolved by deleting some of the files that you no longer need. To determine how many files you should remove, look at the amount of free disk space that is currently available.

Guided Activity 2.5

1. Open your computer's *Start* menu.
2. Enter "file explorer" and click the *File Explorer* option. You should now see a new window open.
3. On the left of the new window, click the *This PC* option. This will open a summary of all the hard drives in your computer.
4. The free available space should now be visible for each of your hard drives. It is displayed as 23.00 Gb free of 118 Gb.

CONNECTION PROBLEMS

If you are struggling to use your computer to connect to the internet, the problem could be caused by any one of the following:

- Your computer
- The network cable
- The router
- The internet cable
- The internet service provider (ISP)

The following table shows you how to identify the cause of the problem.

Table 2.26: *Troubleshooting*

LOCATION	EVIDENCE	SOLUTIONS
Computer	• Other computers on the network are still connected to the internet • The ethernet cable works when plugged into a different computer	• Make sure that your cable or Wi-Fi is connected. • Run the *Troubleshoot Network* application. • Restart your computer.
Network cable	• Other computers on the network are still connected to the internet • Computers using a wireless connection are still connected to the internet • Your computer states that no cable is plugged in • The ethernet cable does not work when plugged into a different computer	• Ensure that the cable is plugged in to your computer and the router properly. • Replace the cable. • Change to a Wi-Fi connection.
Router	• No computer on the network has access to the internet • No computer on the network can access the router's setup page	• Make sure that the router is turned on. • Restart the router. • Unplug all cables from the router, except your own cable. • Connect to the router via Wi-Fi if you were using ethernet, or via ethernet if you were using Wi-Fi.

… *continued*

LOCATION	EVIDENCE	SOLUTIONS
Internet cable	• No computer on the network has access to the internet • You can access the router's setup page • The internet and internet cable light on the router are off	• Plug the internet cable into the router. • Restart the router. • Contact the ISP.
ISP	• No computer on the network has access to the internet • You can access the router's setup page • The internet light on the router is off	• Ensure that you have not reached your data limit. • Restart the router. • Contact the ISP.

While network problems can be caused by a number of different devices, they are often easier to fix than hardware or software problems. Once a network has been set up correctly, most problems can be fixed by simply checking all the connections, restarting the affected computers and restarting the router. If this does not work, most ISPs have excellent support teams that should be able to assist you in resolving the problem.

Activity 2.7

1. Write down the correct answer for each of the following questions.
   a. What application do you use to close a non-responsive program?
      A. *Command Prompt* B. *Task Manager*
      C. *Settings* D. *Disk Manager*
   b. The window of this option displays the available disk space on your computer.
      A. *System* B. *Settings*
      C. *This PC* D. *Search*
   c. Which system settings allow you to change the screen's resolution?
      A. *Power & Sleep* B. *Sound Settings*
      C. *Notifications* D. *Display Settings*
   d. Which of the following will cause your scanned document to be of poor quality?
      A. Inability of the OCR software to read the text
      B. Clean alignment and shiny scanner surface
      C. Poorly attached connection
      D. Document too big for the scanner
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Shortcut used to close non-responding programs	A. Disk fragmentation
2.2 An available resolution option	B. Paper jams
2.3 Causes disk errors	C. *Alt+F4*
2.4 When a piece of paper gets stuck in the printer	D. Papier mâché
2.5 To trace and correct faults in a computer system	E. 1 920 × 1 080
	F. 1 234 × 720
	G. Troubleshooting
	H. Debugging

… continued

Activity 2.7

… continued

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if it is false.
   a. An <u>SSD</u> should be defragmented once in a while.
   b. A lower resolution makes the images and text look <u>smaller</u>.
   c. A lack of disk space can <u>slow down</u> your computer.
   d. <u>SSDs</u> should never be optimised or defragmented.
   e. Having a higher screen resolution can <u>slow down</u> your computer.
4. Answer the following questions:
   a. Give three possible causes to erratic mouse movement and how to fix it.
   b. Describe three of the most common occurring scanning problems and the causes thereof.
   c. What causes a black screen to occur?
   d. You are a learner who has just finished writing a very important essay for an assignment. However, the printer that you are using to print the essay is not working. What could be causing this problem?

UNIT

2.7 New technologies

Computer technology is an ever-changing field. Each day, new technologies are developed to replace or improve existing technologies. The most powerful computing device today will not be the most powerful device tomorrow. This is part of what makes the fields of computers and technology so exciting.

Let's take a look at some examples of technologies that were introduced in 2018 already:

- **Holographic phone:** This opened many new possibilities for communication as images could be viewed in 3D.
- **5G connections:** This type of wireless connection greatly increased the speed of mobile communications, reduced latency and costs, and helped with energy savings.
- **Blockchain technology:** Blockchain technology has been around for several years, with the best known user of this technology being Bitcoin. However, it should be noted that this technology has the potential to change the ways in which we process and store data.

REVISION ACTIVITY

PART 1: MULTIPLE CHOICE

1.1 Which of the following is an example of memory? (1)

A. RAM
B. HDD
C. SSD
D. GPU

1.2 Which of the following is NOT a communication device? (1)

A. Router
B. Switch
C. USB flash
D. Wi-Fi card

1.3 Which of the following networks would a home user use? (1)

A. WAN
B. LAN
C. HAN
D. WLAN

1.4 Which of the following storage is NOT physical? (1)

A. Blu-ray disk
B. ROM
C. Cloud storage
D. Internal hard drive

1.5 Which of these problems has to do with the computer's software? (1)

A. The mouse is not responding
B. The computer is overheating
C. The computer cannot connect to the printer
D. The computer keeps freezing when you run a program

[5]

PART 2: TRUE OR FALSE

Indicate if the following statements are TRUE or FALSE. Correct the statement if it is false. Change the underlined word(s) to make the statement true.

2.1 <u>SOHO users</u> are people who work at home and in small offices. (1)

2.2 A smartphone is ideal for a <u>mobile user</u>. (1)

… *continued*

REVISION ACTIVITY

... continued

2.3 When you use a keyboard, make sure that your elbow does not bend a lot. (1)
2.4 A digital camera captures photographs and stores them on an internal hard drive. (1)
2.5 Cache is located on the hard disk, while RAM is located on the CPU. (1)
[5]

PART 3: MATCHING ITEMS

Choose a term or concept from Column B that matches a description in Column A. (5)

COLUMN A	COLUMN B
3.1 Over the years, Mary-Anne's computer has started performing slower	**A.** Input device **B.** Output device
3.2 Kabelo uses loud speakers to listen to his favourite music	**C.** Network error
3.3 Patrice saves all her college assignments on her flash drive so that she can access them with any computer	**D.** Processing device **E.** Printer error
3.4 Sangram uses a router to connect his smartphone to his laptop and tablet	**F.** Communication device **G.** Storage device **H.** Display screen error
3.5 Busisiwe's computer keeps disconnecting from the internet every ten minutes	**I.** Embedded device

[5]

PART 4: CATEGORISATION QUESTIONS

Classify the following pieces of hardware as input, processing, output, or communication devices. (10)

HARDWARE	TYPE OF DEVICE
4.1 Touch screen	
4.2 Headphones	
4.3 Printer	
4.4 GPU	
4.5 SSD	
4.6 Microphone	
4.7 Modem	
4.8 CD/DVD writer	
4.9 Scanner	
4.10 Router	

[10]

PART 5: MEDIUM QUESTIONS

5.1 Provide three examples of an input device that a paralysed person would use. (3)
5.2 How do you access and use the *Optimise Drives* tool? (5)
5.3 If you want to watch a video on YouTube, what hardware do you need? Mention ONE piece of hardware you will need for each step of the information-processing cycle. (5)
5.4 Which do you think is better, wired or wireless input devices? Give a reason for your answer. (2)
[15]

TOTAL: [40]

AT THE END OF THE CHAPTER

NO.	CAN YOU ...	YES	NO
1.	Evaluate hardware devices?		
2.	Suggest input, output, storage and communication devices, as well as CPU and RAM; including specifying basic specifications in terms of processor, memory and storage for: • home users • SOHO users • mobile users • power users • disabled users?		
3.	Fix ordinary hardware problems?		

TERM 1

CHAPTER 3

SOFTWARE

CHAPTER OVERVIEW	
Unit 3.1	Uses of common applications
Unit 3.2	Software enhancing accessibility, efficiency and productivity
Unit 3.3	Interpreting system requirements
Unit 3.4	Common software problems
Unit 3.5	Social implications: User-centred design

By the end of this chapter, you will be able to:

- Explain why software is important.
- Identify some of the most commonly used software applications.
- Describe how software can enhance accessibility, efficiency and productivity.
- Understand the difference between web-based and installed applications.
- Interpret software system requirements.
- Describe some of the most common software problems.
- Explain the risk of using flawed software.
- Discuss user-centred design.
- Understand the social implication of software design.

INTRODUCTION

When you think of a computer, the first thing that comes to mind will probably be the physical components (the hardware). However, a computer would not be able to function without something giving instructions to the hardware. This "something" is called **software**. Software is a collection of computer instructions that tells the computer how to perform specific tasks. Software, in other words, tells a computer how to work.

Something to know

Unlike the hardware components discussed up to this point (which have all been physical devices that you can hold in your hands), software is not a physical device that you can connect to the computer with cables. Software is stored on your computer's storage device and it tells the different hardware devices how to handle the input they receive.

As a result, hardware and software are said to be **interdependent**, which means neither of them can exist without the other one. Hardware cannot do anything without the software telling it what to do and software cannot exist without the hardware on which it runs.

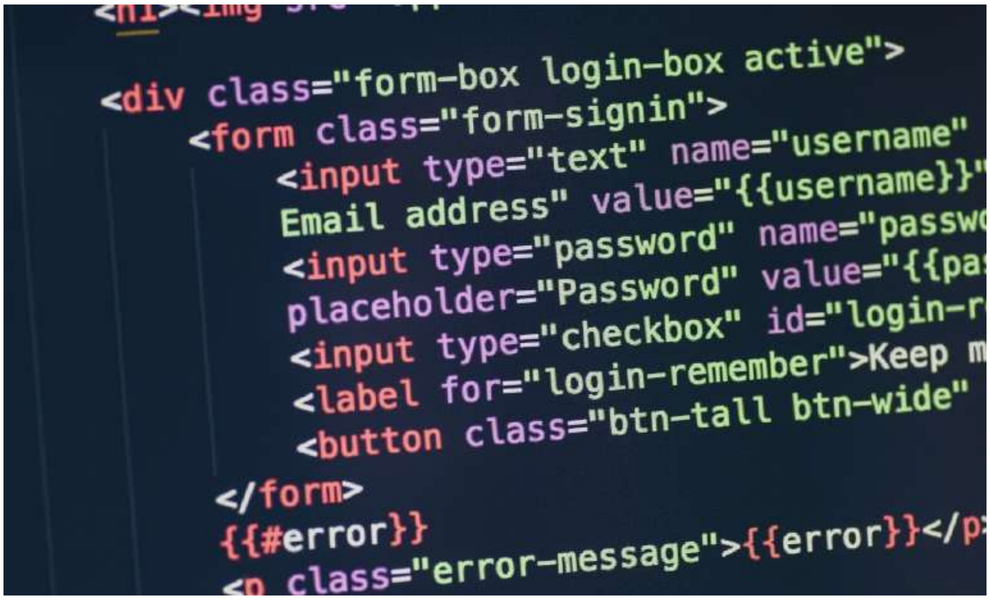


Figure 3.1: *Software is the instructions that a programmer has created for a computer*

To better illustrate this, let's look at the following scenario. Computer hardware is like a team of workers that can do many different jobs – for example, dig holes, mix cement, join wood, lay bricks, fit glass, pour a foundation, build a roof, and so on. The computer software is the lead builder, who tells each worker exactly what to do, where to do it and when to do it in order to build the house.

In this chapter, we will take a closer look at some of the most commonly used computer applications, as well as software that enhances your productivity and efficiency. We will also discuss the most common software problems and why it is risky to use faulty software.

UNIT

3.1 Uses of common applications

Application software is a type of software that allows the user to perform a specific personal, educational, or business-related function. In short, application software refers to the computer programs that we use on a daily basis. Examples of these include the following:

- Word processing
- Spreadsheet
- Database
- Presentation
- Email
- Document management
- Web browsers

WORD-PROCESSING SOFTWARE

Word-processing software is a type of program that allows the user to compose, edit, format, save and print typed documents. Let's take a look at some of the other features of word-processing software:

- **Graphics:** Word processors contain a variety of different backgrounds, clipart and colours that can be used. It also allows you to add images, photos and videos obtained from an external source.
- **Templates:** Word processors possess the ability to create templates that can be used to standardise documents. This is extremely useful if you have to write multiple documents that cover the same topics.
- **Spell checker:** Word processors come equipped with a built-in spell checker that will check your spelling and grammar. You can update the spell-checker's database to include words that are not used very often, for example scientific names, such as *Beauveria bassiana*.
- **Thesaurus:** Word processors come equipped with a built-in thesaurus that will suggest similar words to the word that you are currently using.

If you are writing a letter, doing an assignment, or making a few notes, there are a few word-processing software applications that you can use. Table 3.1 shows examples of these.

Table 3.1: *Types of word-processing software*

PAID	FREE	ONLINE	
Microsoft Word	LibreOffice	Google Docs	Office 365

INSTALLED APPLICATIONS VERSUS WEB-BASED APPLICATIONS

Installed applications are applications that can be accessed on your computer without the need of an internet connection. Installed applications refers to software that is permanently installed onto the hard drive of the computer

The following table lists some of the advantages and disadvantages of using installed software.

Table 3.2: *The advantages and disadvantages of using installed software*

ADVANTAGES	DISADVANTAGES
No internet connection is needed	The user must install the application on the computer before using it
Installed applications offer users with more features and functions	When the user installs the software on one computer and he or she can only access it on that computer
Data is stored on the local disk of the computer	User is responsible for backing up and saving data
Allows for faster data entry and reporting as data does not need to be uploaded to the internet	Applications take up storage space on the user's computer, which results into less disk space
	Software might need to be updated, which requires an internet connection

Web-based applications are stored on servers on the Internet or an Intranet. Users need an internet connection to access the software.

Table 3.3: *The advantages and disadvantages of using web-based software*

ADVANTAGES	DISADVANTAGES
Software does not take up any disk space on the computer	Requires an internet connection
Data is managed and backed up by the software providers	Application might be slower than an installed application due to internet speed
Can be used anywhere, provided one has an internet connection	Interfaces of web-based applications are not always as highly developed as those on installed applications
In large organisations, all the computers with a web browser can access the application without any individual installations being needed	Data is not always safe and secure on the internet
Users do not need to update the application on their computer because it will be updated on the server where the application is hosted	The web browser needs to be compatible with the web-based application

When trying to decide whether to use installed or web-based programs, look at your personal needs and situation. For example, if you do *not* have an active internet connection, web-based applications will not work.

SPREADSHEET SOFTWARE

Spreadsheet software is a type of program that sorts, arranges and analyses data in a table format. The user enters data into the columns and then the software performs various calculations.

Let's take a look at some of the other features of spreadsheet software:

- **Forms:** The user can create forms that can be used for a variety of applications, including time sheets, surveys and review forms.
- **Data analysis:** The user can analyse the data that is entered into the workbook cells. This can be done using a variety of formulae available in the workbook, or through graphs (which can be created automatically with the *Insert Graph* function).
- **Conditional formatting:** The user can customise the data based on the content of the cells in the workbook. This can be used to highlight errors, identify important patterns in data, or to keep track of activities (for example, to show which employees have not been signing in with their time sheets).
- **Sorting and filtering:** Spreadsheets may contain a huge amount of data. To make it easier to find the data you need, spreadsheet software can sort and filter the data, based on your requirements.

Figure 3.2: *Spreadsheets*

The following table lists some examples of spreadsheet software.

Table 3.4: *Types of spreadsheet software*

PAID	FREE	ONLINE
Microsoft Excel	LibreOffice	Google Sheets

DATABASE SOFTWARE

Database software is designed to create and manage databases. A database is an organised collection of data, usually stored in the form of structured fields, tables and columns. With database software, the user can create, edit and maintain the database files, and sort, search for and retrieve information when needed.

Let's look at some of the other features of database software:

- **Security:** One of the biggest concerns when working with data is how secure it is. With database software, a user can protect his or her data by giving only specific people access to the data.
- **Permanent storage:** Database software makes it possible to create a database that can store data permanently. To help with this, the software is equipped with a file backup and recovery system.

The following table shows some examples of database software.

Table 3.5: *Types of database software*

PAID	FREE	ONLINE
Microsoft Access	MySQL	Amazon SimpleDB
Microsoft SQL Server		

PRESENTATION SOFTWARE

Presentation software is specifically designed to help the user create and edit presentations in the form of a slide show. Presentations can include text, videos, or images.

Let's take a look at some of the features of presentation software:

- **Animation effects:** Animation effects allow the user to add animations to the text and images on the slides. This helps to give the presentation a more interactive feel.
- **Slide notes:** Slide notes allow the user to add notes and comments to each slide. This is especially useful to add a description for an image or video.
- **Transitions:** Transitions allow the user to customise how the slides change when going from one slide to the next. This can help make the presentation "feel" like it is a video.

Figure 3.3: *Presentation templates*

The following table lists some of the examples of possible presentation software.

Table 3.6: *Types of presentation software*

PAID	FREE	ONLINE
Microsoft PowerPoint	Apache OpenOffice Impress	Google Slides

EMAIL SOFTWARE

Email software makes it possible for a user to compose and send email messages to other people using the internet, as well as to receive such messages. Email messages can include text, videos and images.

Let's take a look at some of the other features of email software:

- **Address book:** The address book allows the user to store and sort the email addresses of friends, family and work colleagues.
- **Mailing list:** This makes it possible to send one email to multiple recipients.

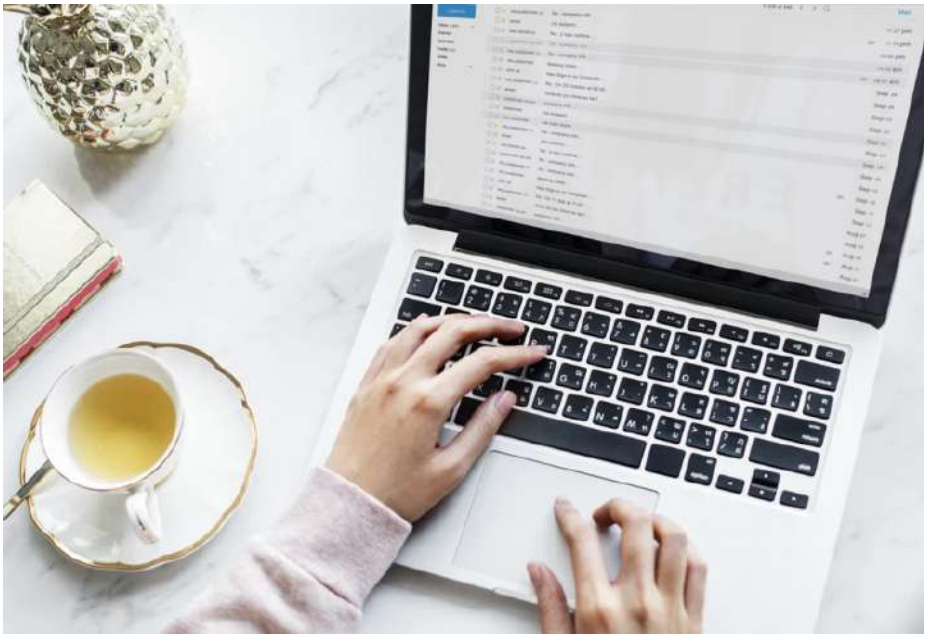


Figure 3.4: *Using email software*

The following table lists some of the examples of email software.

Table 3.7: *Types of email software*

YAHOO	OUTLOOK	GMAIL
Y! YAHOO! MAIL	Outlook	

DOCUMENT MANAGEMENT SOFTWARE

Document management software is a type of program that allows the user to store, manage, and track electronic documents and images. One of the ways in which this is done is by converting the document to a PDF format, as it prevents the user or recipients from mistakenly making any changes to the document.

Let's take a look at some of the other features of document management software:

- **Password protection:** This allows you to encrypt and protect your documents. Only people who have the correct password will be able to access and read the documents.
- **Restricted access:** Restricting the access to your documents allows only certain people to view specific documents. As with password protection, this helps to protect your documents.
- **Version control:** The software helps maintain version control by enabling you to see all the versions of a specific document, including its latest version.

Figure 3.5: *Organising documents through software*

WEB BROWSERS

A web browser is a software application that allows the user to access information on the internet. This is done by allowing the user to open and view web pages.

Examples of web browsers include the following:

- **Microsoft Edge:** This web browser comes pre-installed with any computer that runs on Windows 10.
- **Google Chrome:** This is a free web browser developed by Google.
- **Firefox:** This is a free open-source web browser developed by Mozilla.

Figure 3.6: *Types of web browsers*

Activity 3.1

1. Write down the correct answer for each of the following questions.
   a. Which of the following are some of the features of email software?
      A. Accessibility, multimedia, hyperlinks, updated
      B. Password protection, restricted areas, version control
      C. Instant messaging, address book, antivirus, mailing list
      D. Security, permanent storage
   b. Which of the following are some of the features of spreadsheet software?
      A. Instant messaging, address book, antivirus, mailing list
      B. Forms, data analysis, conditional formatting, sorting and filtering
      C. Graphics, templates, spell checker, thesaurus
      D. Accessibility, multimedia, hyperlinks, updated
   c. Which of the following are some of the features of presentation software?
      A. Graphics, templates, spell checker, thesaurus
      B. Password protection, restricted areas, version control
      C. Accessibility, multimedia, hyperlinks, updated
      D. Animation effects, slide notes, transitions
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Software you can access over the internet	A. Database software
2.2 Software that sorts, arranges and analyses data in a table format	B. Email software
2.3 Software that is designed to create and manage databases	C. Presentation software
2.4 Software that allows the user to store, manage and track electronic documents and images	D. Spreadsheet software
2.5 Software that allows the user to compose, edit, format, save and print typed documents	E. Web-based application software
2.6 Software that is specifically designed to help the user create and edit presentations in the form of a slide show	F. Word-processing software
2.7 Software that makes it possible for a user to compose, send and receive email messages to and from other people using the internet	G. Document management software

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if they are false.
   a. Firefox is a web-browser developed by <u>Microsoft</u>.
   b. Wikipedia is a free, online <u>email software</u>.
   c. <u>Google Maps</u> is a free reference software, but it cannot work without an internet connection.
   d. The serial key required for activating installed software is legally and freely given on the software application's website.
   e. One of the biggest concerns when working with data is the <u>internet connection</u>.

… continued

Activity 3.1 *… continued*

4. Answer the following questions:
   a. What advantages do web-based applications have over installed applications?
   b. A big disadvantage of web-based software is that an internet connection is required. Is this a limiting factor for most South Africans? Explain why.
   c. List one example of each type of free software.
   d. Name and explain four features of word-processing software, and give instructions on the necessary steps that have to be taken in order to use these features.
   e. While installing software, you were not required to enter a product key. Give a reason as to why this occurred.

UNIT

3.2 Software enhancing accessibility, efficiency and productivity

In previous sections, we looked at why software is important. We also looked at the most commonly used programs. However, the usefulness of software is not just limited to you being able to use your computer. Software can also play an important role in helping to increase your efficiency and productivity, as well as improve the lives of people with disabilities.

To better illustrate this, we will look at the following examples of software that make our lives easier:

- Voice-recognition software
- Typing tutor or keyboarding skills
- Note-taking software
- Cloud applications

HOW VOICE RECOGNITION WORKS | CADILLAC

https://www.youtube.com/watch?v=cSsnYmopB8M

VOICE-RECOGNITION SOFTWARE

Voice-recognition or speech-recognition software is a type of program that allows the computer to take verbal commands from the user and then interpret them. It does this by converting the audio that is received from the microphone, to digital signals that the computer can understand. The signals are then compared to a database containing words and phrases, as well as the actions that should be performed.

Voice recognition has many advantages that help to make our daily lives better, including the following:

- Helping people with physical disabilities to use a computer
- Giving the option to make phone calls without having to touch a phone (which is very useful when driving especially)
- Using the software to search for information on the internet, for example using Siri or Alexa to search for a recipe while cooking
- Improving productivity as you can talk much faster than you can type or write
- Improving security, as your voice can be used as a security measure to help protect your data

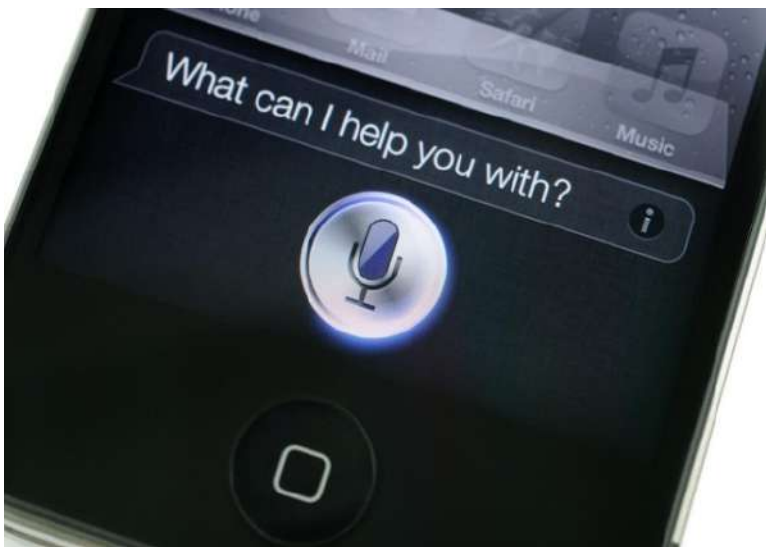


Figure 3.7: *Voice-recognition software installed on a smartphone*

The following table lists some examples of voice-recognition software.

Table 3.8: *Types of voice-recognition software*

PAID	FREE	ONLINE
Dragon Professional Individual	Windows 10 Speech recognition (this is included with the Windows operating system) Google Docs Voice Typing	Google Docs Voice Typing

TYPING TUTOR OR KEYBOARDING SKILLS

A **typing tutor** is a type of software that teaches a user how to use the keyboard more effectively and accurately, as well as improve typing stamina and speed. The program supplies exercises in which the user must type certain words and phrases within a specific timeframe.

In order to make these activities more fun and interactive, many typing tutor programs have been designed in the form of a game. An example of this is the game "Typing War Master". In this game, the objective is to protect your town by typing the words contained within the bombs falling from the sky. If you type the word correctly, the bomb disappears. If you mistype a word, the bomb falls and destroys a part of your town.

The following table lists some examples of typing tutors.

Table 3.9: *Types of typing tutors*

PAID	FREE	ONLINE
Typesy	RataType	Speed Typing Online

NOTE-TAKING SOFTWARE

Much like taking notes on a notepad, note-taking software allows the user to take digital notes on the computer. The notes are then recorded, organised and stored in a single place, allowing the user to search for specific documents without any hassle.

Note-taking software is not limited to just text; you can include images and videos. Let's take a look at some of the advantages of note-taking software:

- Your notes are stored on your computer, preventing them from getting lost.
- Your notes are always at hand and can be accessed as long as you have your smartphone or computer with you. There is also no risk of running out of pages on which to write.
- Note-taking software enables you to search for specific topics and information. This makes it much easier for you to work through the notes.

Figure 3.8: *Note-taking software installed on a tablet*

The following table lists some examples of note-taking software.

Table 3.10: *Types of note-taking software*

PAID	FREE	ONLINE
Evernote Microsoft OneNote	Evernote (can upload 60 Mb free per month)	Simplenote

CLOUD APPLICATIONS

Cloud computing is a new technology that allows the user to store information, or use applications on the internet, instead of being on his or her own computer. It uses a network of remote servers to maintain data storage and software applications so that the user can access the service from any device that can connect to the internet.

Let's take a look at the following examples of cloud computing applications:

- Email services, such as Gmail
- Cloud storage, such as OneDrive and Dropbox

The main advantage of cloud computing is that it allows the user to store large amounts of data and information, without the worry of running out of hard-drive space, or having to purchase expensive software applications.

The advantages of cloud computing are:

- It saves the cost of maintaining your own IT equipment.
- It is available 24/7.
- It offers backup and recovery functions.
- It uses automatic software integration.

The disadvantages are:

- Sometimes users experience technical issues.
- Security concerns in the cloud.
- It could be prone to attacks.
- Need internet connection.

Activity 3.2

1. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
1.1 A type of program that allows the computer to interpret voice commands from the user	A. Note-taking software
1.2 A software where the user can take digital notes on the computer	B. Simplenote
1.3 An online note-taking software	C. Voice-recognition software
1.4 An online voice-recognition software	D. Dragon Professional Individual
1.5 A note-taking software for which you pay	E. Microsoft OneNote
	F. Evernote
	G. Notepad
	H. Google Docs voice typing

2. Answer the following questions:
   a. Give three advantages of note-taking software.
   b. How does note-taking software help the environment?
   c. Give three advantages of voice-recognition software.
   d. Name an example of free and paid-for voice-recognition software.
   e. One of the advantages of voice-recognition software is that it increases productivity. Explain through the use of an example why this is so.

UNIT

3.3 Interpreting system requirements

Something to know

System requirements are guidelines; not absolute rules. If your hardware does not meet the system requirements, it does not mean that the software will not work. However, there is a good chance that the software will perform very poorly and might stop working in the middle of the process. It is, therefore, recommended that you always try and meet at least the minimum requirements.

By now you know that all computers are made up of hardware and software. A computer will not function properly if it receives instructions from software that it cannot carry out. It is, therefore, very important that you get the correct software for your hardware.

SYSTEM REQUIREMENTS OF SOFTWARE

In order to determine which software would work best for your computer, you will need to look at the **system requirements** of the software. The system requirements refer to a list of hardware and software requirements that the software will need in order to function. The list will normally include the **minimum** (the lowest specifications needed to let the program work) and **recommended** (the specifications where the program should function best) hardware requirements for the software. This includes the following:

- The amount of hard-drive space needed to install the software
- The amount of RAM needed for the software to function normally
- The GPU and CPU required to do calculations
- The type of programs (usually the type of operating system) that is required for the new program to function
- The input and output device needed for data to be collected and displayed

The following table is an example of the system requirements that software might need in order to function.

Table 3.11: *System requirements to play a game called "World of Warcraft"*

COMPONENTS	MINIMUM	RECOMMENDED
Operating system	Windows 7 64-bit	Windows 10 64-bit
Processor	Intel® Core™ i5-760, or AMD FX™-8100, or better	Intel® Core™ i7-4770, or AMD FX™-8310, or better
Video	NVIDIA® GeForce® GTX 560 2 Gb or AMD™ Radeon™ HD 7850 2 Gb or Intel® HD Graphics 530 (45W TDP)	NVIDIA® GeForce® GTX 960 4 Gb or AMD™ Radeon™ R9 280, or better
Memory	4 Gb RAM (8 Gb for Intel HD Graphics 530)	8 Gb RAM
Storage	70 Gb available space 7 200 RPM HDD	70 Gb available space SSD
Internet	Broadband internet connection	Broadband internet connection
Input	Keyboard and mouse required; other input devices are not supported	Multi-button mouse with scroll wheel
Resolution	1 024 × 768 minimum display resolution	1 024 × 768 minimum display resolution

The following table is an example of the basic system requirements for entry-level computers – for personal or SOHO users.

GENERAL SYSTEM REQUIREMENTS FOR PC (E.G. STUDENT LAPTOP)	SOFTWARE FOR PC
• Intel Core i3 processor • 4 GB RAM • 500 GB HDD • Wireless Ethernet (802.11 g/n) • Windows 10 • Microphone and camera	• **MS Office for PC 2013 to 2019 or Office 365** • **Adobe Reader (PDF Reader):** Many of the documents shared with students are in .pdf format. You will need this software to download and read these documents. The program can be downloaded for free at: www.adobe.com • **Oracle Java Runtime:** Oracle Java Runtime is a free web-browser plug-in that enables interactive media experiences, rich business applications and immersive mobile apps. The system requirements for Oracle Java Runtime and the following associated technologies: (Windows Operating System: Windows 7, Windows 8, Windows 10) (Intel® Pentium® III 450MHz or faster processor or equivalent) (4GB of RAM) at www.java.com. *Note: Most browsers require that Java be enabled in the preferences.* • **Adobe Shockwave:** Adobe Shockwave is a free web-browser plug-in that enables interactive media experiences, rich business applications and immersive mobile apps. The system requirements for Adobe Shockwave and the following associated technologies: (Windows Operating System: Windows 7, Windows 8, Windows 10) (Intel® Pentium® III 450MHz or faster processor or equivalent) (4GB of RAM) at www.adobe.com

Activity 3.3

1. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if they are false.
   a. If your <u>software</u> does not meet the system requirements, it means that the software will not work.
   b. Running out of <u>RAM</u> may cause programs to stop working.
   c. Certain programs cannot work if you do not have the right <u>operating system</u>.
   d. <u>Modern laptops</u> (after 2009) meet systems requirements for Office packages (for example, Microsoft Word, Microsoft Excel and Microsoft PowerPoint).
   e. A computer will not function properly if it contains <u>software</u> instructions that cannot be performed by the hardware.
2. You have recently purchased the Dell laptop described below.
   The requirements to run Windows 10, an image editing software and two games are given.
   Use the information given to answer the questions that follow.

Minimum requirements, *Call of Duty 4:* Modern Warfare 2	**Recommended requirements, *Call of Duty 4*:** Modern Warfare 2
OS: Win 7 × 64	OS: Win 10 × 64
Processor: Intel Core i3-530 2.9 GHz / AMD Phenom II X4 810	Processor: Intel Core 2 Duo E8500 3.16 GHz / AMD Athlon II X3 415e
Graphics: AMD Radeon HD 7850 or NVIDIA GeForce GTS 450 v4	Graphics: AMD Radeon R7 265 or NVIDIA GeForce GTX 750 Ti
VRAM: 1 Gb	VRAM: 2 Gb
System Memory: 6 Gb RAM	System memory: 6 Gb RAM
Storage: 55 Gb hard-drive space	Storage: 55 Gb hard-drive space
DirectX 11 compatible graphics card	

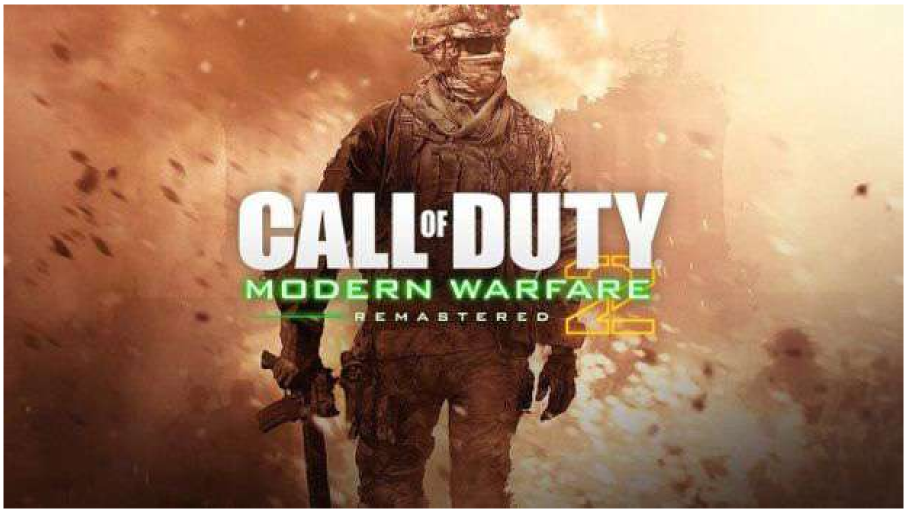


Figure 3.9: *A scene from "Call of Duty 4"*

… continued

Activity 3.3

… *continued*

a. What difference does it make to your computer when you run a program that makes the minimum requirement, compared to a program that is run at the recommended requirements?
b. Does the Dell laptop have the minimum requirements for Windows 10? Show how you came to this answer.
c. What does OS mean in the "Call of Duty" game requirements?
d. What does display resolution 1 366 × 768 mean?
e. What is meant by "touch display"?
f. Give an example of an image editing software program.
g. Can the "Call of Duty: Modern Warfare 2" and image editing software be installed on the laptop?
h. List the different processors' names in the advertisements and sort them from fastest to slowest.

UNIT

3.4 Common software problems

Something to know

It is very important to install program updates, as it keeps your program running safely and efficiently. It also prevents others from exploiting your computer, as cybercriminals tend to target older, flawed versions of software that contain unpatched security flaws.

Something to know

Developers are aware that you might not always be able to download every update as they come out. To help with this, they might bundle together a number of update patches into a service pack. Downloading this service pack will then update your software with all the missed patches. However, it is still recommended that you update your software regularly. The service pack is there to help people who might not have access to the internet on a regular basis.

As with most things in life, software comes with its own problems. Programmers often make small errors when they write the code for software. The process is further complicated by the fact that most software is designed to be used by a variety of different users, on many different computers and operating systems.

UPDATING SOFTWARE

It is very difficult for developers to test the software for each possible mistake that might occur. Because of this, software developers use software updates to fix any problems that have been identified once the software has been released. The updates also include improvements to program performance, stability and security, as well as new program features.

Software developers regularly release updates in the form of update patches. These patches can vary in size, depending on the amount of new program information contained within the patch. The patches can be downloaded from the developer's website. However, most programs have a setting that automatically updates the program whenever an update is available. This setting can be changed so that it asks your permission to update the program. This is especially useful if you have limited bandwidth.

READ-ONLY FILES

Read-only files allow users to open and read the file, but no changes can be made to the file. This protects data from being changes on purpose or by accident.

If you find that you cannot save or edit a file, it may be that the file had accidentally been changed to "read-only". This problem might occur due to software updates that contain a faulty code. Despite this, software developers are quick to release updates to fix this issue.

The term **Software versioning** is defined as a way to categorise the unique levels of computer software as it is developed and released. The identifier can be a number, a word, or both. For example, version 1.0 is commonly used to denote the initial release of a program.

A **software upgrade** is a new version of the software that offers a major improvement over the current version of your software. In many cases, a software upgrade requires you to purchase the new version of the software, sometimes at a discounted price if you own an older version of the software.

Guided Activity 3.1

Follow these steps to turn off the read-only setting:

1. Go to the file you are trying to edit.
2. Right click the file and select *Properties* from the options list. This will open a new *Properties* window.
3. At the bottom of the *Properties* window, under *Attributes*, cancel the *Read-only* option selection.
4. Click *OK* at the bottom of the *Properties* window.

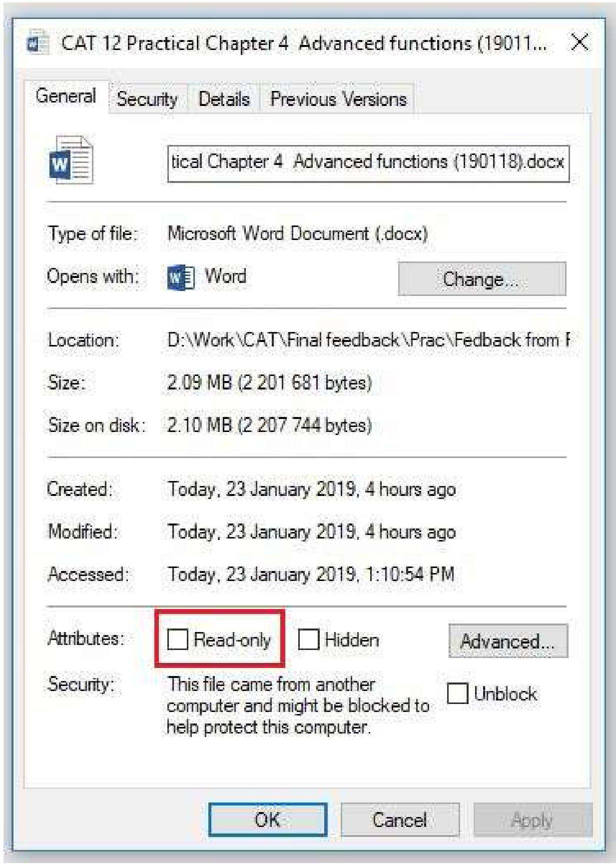


Figure 3.10: *Turning off the "Read-only" setting*

Once you have turned off the read-only attribute of the file, you should be able to edit and save the file as normal.

RISKS OF USING FLAWED SOFTWARE

As we have stated previously, not all software is perfect. Some software might contain flaws or bugs that will cause the system to produce an incorrect or unexpected result. Using such software has certain risks associated with it. These risks include the following:

- **Security risks:** Faulty software might not be fully secure and can contain flaws that people with malicious intent can exploit in order to access your computer and steal your data.
- **Underperformance:** Faulty software might cause your computer to underperform due to slow system response and transaction rates.
- **No performance:** Faulty software can lead to program and computer crashes. This might require you to restart your computer.

Something to know

Software developers use read-only files by selling their software on CDs/DVDs. This allows you to install the program onto your computer, but will prevent you from making changes to the installation files on the CD/DVD. That way, even if you make changes to the program on your computer, you will always have the CD/DVD containing the original program.

- **Navigation risks:** Faulty software can cause programs that use a global positioning system (GPS) to give inaccurate results. This can lead to the user getting lost and wasting a lot of time trying to determine how to get to his or her destination.
- **Economical risks:** Faulty software used to make payments to workers can lead to some workers getting paid more than intended, while others might not get paid at all.

When you encounter a fault or bug in software, you should report the fault to the software developers. This can be done on the software's Home page. Make sure to include as much information as possible, including what the error is, how it occurred and how often it occurs. By letting the developers know of the fault, you will help to ensure that all users have a fault-free program to use.

Activity 3.4

1. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if they are false.
   a. Software updates include improvements to program performance, stability and <u>improved access.</u>
   b. Software developers use <u>write-only</u> files by selling their software on CDs/DVDs.
   c. <u>Faulty software</u> leaves space for people with malicious intent to exploit the user.
   d. Flawed software can lead to problems, such as security risks, underperformance and <u>overperformance</u>.
2. Answer the following questions:
   a. What is the difference between a service pack and a patch?
   b. How do read-only files work?
   c. Is it possible for a home user to change a file's attribute to or from "Read-only"? If so, explain how this can be done.
   d. Name and describe five risks associated with using flawed software.
   e. How can you minimise the risk of faulty or flawed software?
3. Bongi has recently downloaded a music album. She has a certain way of naming her music files, but is having trouble in trying to rename them.
   a. Why do you think Bongi is having this problem? Name at least two possible causes.
   b. Explain in detail how Bongi should go about fixing the problems mentioned in your previous answer.

3.5 Social implications: User-centred design

User-centred design (UCD) is a design process whereby the developers design software based on the focus and needs of the users. This involves a four-phase process:

1. Identify how the user will use the product
2. Determine the user's requirements
3. Design the product
4. Evaluate the product according to the user's needs, requirements and product performance

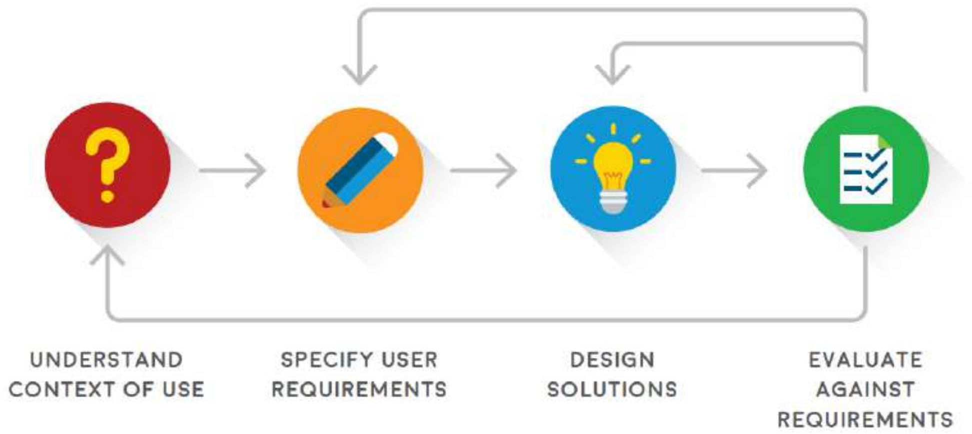


Figure 3.11: *The process of user-centred design*

Based on the evaluation of the product, the whole process might be moved back one step, or even restarted from the beginning.

Advantages of UCD include the following:

- **Meeting customers' expectations:** This will lead to an increase in sales and lower costs caused by returns and customer support.
- **Safer products:** Because UCD focuses on the user, products can be designed for specific tasks. This reduces the risk that the product will be used for the wrong task.
- **No training:** UCD allows products to be designed in a user-friendly manner. This reduces the amount of training needed to use the product and ensures that customers are not frustrated by using it.
- **Designing websites:** UCD plays an important role when designing a new website, as users need to be able to navigate and find what they are looking for without having any problems. The following factors should be considered when designing a website:
  - Visibility
  - Accessibility
  - Legibility
  - Language

Figure 3.12: *An example of a bad website design*

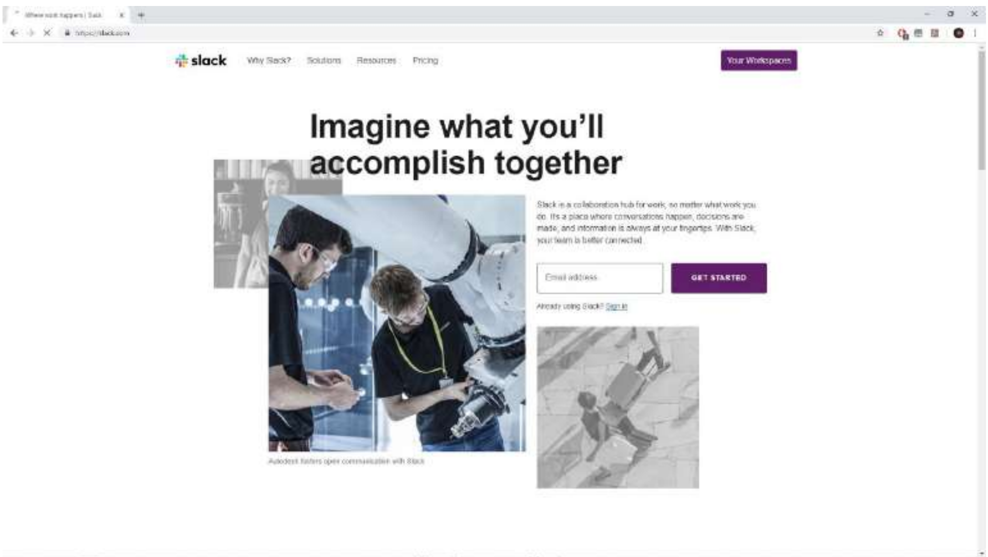


Figure 3.13: *An example of a good website design*

- UCD makes it easier for a user to enter data into a database, as it ensures that the user enters the correct information where needed. The following factors should be considered when designing a database:
  - Purpose of the database
  - Type of data being stored
  - Data fields required for data entry
  - Data-validation techniques to protect data entry
- UCD assists in making presentations easier to present and understand. This can be done in the following ways:
  - Keep bullet points and text to a minimum
  - Avoid using too many transitions
  - Use high-quality images
  - Use the white space (blank space); avoid cluttering the slide with too much information and unnecessary images

Figure 3.14: *An example of a bad presentation slide*

Figure 3.15: *An example of a good presentation slide*

Activity 3.5

1. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
1. Third step in the UCD process	A. No training needed
2. A factor to be considered when designing a website	B. Understand context of use
3. One of the advantages of using UCD	C. Yes
4. Are visuals an important factor in website design?	D. No
5. The step in UCD that has input in the first step of UCD	E. Accessibility
	F. Design method
	G. Design solutions
	H. Evaluate against requirements

… continued

Activity 3.5

... continued

2. Answer the following questions:
   a. You are tasked to create a presentation about your favourite game. Which process will you use to design and create the presentation? Explain how this process works.
   b. Make a free-hand drawing to better explain the process you described in the previous question.
   c. Is this process something you do once off, or something that should be done on a continuous basis?
   d. Will you be able to use the same process when designing and creating a new website?

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

1.1 Which of the following is an example of online spreadsheet software? (1)
   A. Google Docs
   B. Google Sheets
   C. Microsoft Excel
   D. Microsoft PowerPoint

1.2 Which of the following is a limitation of installed software? (1)
   A. You must back up and save your data
   B. Data is stored on your computer
   C. You need an internet connection to access your data
   D. It is slower than web-based software

1.3 Which of the following has a built-in antivirus? (1)
   A. Email software
   B. Web browsers
   C. Database software
   D. Spreadsheet software

1.4 Which of the following would be the output equivalent of voice-recognition software when used by blind users? (1)
   A. Eye-motion sensors
   B. Text-to-speech software
   C. Braille interface
   D. Gesture-recognition software

1.5 Which of the following is NOT an advantage of UCD? (1)
   A. Produces safer products
   B. Requires training
   C. Meets customer expectations
   D. Improves website layout

[5]

QUESTION 2: TRUE OR FALSE

Indicate if the following statements are TRUE or FALSE. Correct the statement if it is false. Change the underlined word(s) to make the statement true.

2.1 Software tells the computer's <u>hardware</u> what to do. (1)
2.2 <u>Reference software</u> uses templates. (1)
2.3 You will always have access to <u>installed software</u> when compared to <u>web-based software</u>. (1)
2.4 <u>Word-processing software</u> and reference software use <u>dictionaries</u>. (1)
2.5 PDFs are used by <u>word-processing software</u> to restrict user access to documents. (1)
[5]

... continued

REVISION ACTIVITY

… continued

QUESTION 3: MATCHING ITEMS

Choose a term or concept from Column B that matches a description in Column A. (5)

COLUMN A	COLUMN B
3.1 A secure way of storing your data	A. Read-only files B. Document management software C. Presentation software D. Typing tutor E. Minimum system requirements F. Cloud applications G. Recommended system requirements H. Updating software
3.2 Software that allows you to add animation effects to text	
3.3 A document that does not allow you to edit, rename, or save changes to it	
3.4 The system requirements needed to make a program work	
3.5 Software that makes use of **encryption** and passwords to protect your information	

[5]

QUESTION 4: CATEGORISATION QUESTIONS

Using the following table, determine which applications are used for the following software. Copy and complete the table. (8)

- Word-processing software
- Spreadsheet software
- Database software
- Presentation software
- Reference software
- Email software
- Web browsers
- Document management software

SOFTWARE	APPLICATION
4.1 Microsoft PowerPoint	
4.2 Google Maps	
4.3 FileInvite	
4.4 Opera	
4.5 MySQL	
4.6 Microsoft Outlook	
4.7 Wikipedia	
4.8 Microsoft Excel	

[8]

… continued

REVISION ACTIVITY *... continued*

QUESTION 5: SCENARIO-BASED QUESTIONS

5.1 Recently, Sizwe's computer programs have been running very slowly and providing him with constant error messages. Since he bought his computer two years ago, he has never allowed his computer to access the internet.

a. What kind of software problem do you think Sizwe has? (1)

b. Give two reasons why it is important for Sizwe to address this problem? (2)

c. Now that Sizwe has started to access the internet more regularly, he has noticed that some websites do not work on his current web browser. Mention two things that could be causing this problem. (2)

d. Sizwe decides to update all the software on his computer, but some of them do not want to update. Give two reasons why this might be happening and provide a solution to each problem. (4)

e. Explain to Sizwe the difference between minimum requirements and recommended requirements. (4)

5.2 The Tinder dating app revolutionised online dating, so much so that it made a social-cultural impact.

a. Do you think that the creator of the app used the UCD process when creating the app? Give two reasons for your answer. (2)

b. Provide two advantages of using the UCD process. (2)

[17]

TOTAL: [40]

AT THE END OF THE CHAPTER

NO.	CAN YOU ...	YES	NO
1.	Explain why software is important?		
2.	Identify some of the most commonly used software applications?		
3.	Describe how software can enhance accessibility, efficiency and productivity?		
4.	Explain the difference between web-based and installed applications?		
5.	Interpret software system requirements?		
6.	Describe some of the most common software problems?		
7.	Explain the risk of using flawed software?		
8.	Discuss UCD?		
9.	Explain the social implication of software design?		